

# Blind worksheet — model results companion

48 beats, SAME order as clinical\_blind.pdf

Use AFTER the cardiologist has marked her sheet.

Each page corresponds to the matching beat number in clinical\_blind.pdf and shows:

- the same 2-second window in z-scored space
- the model's reconstruction (orange overlay)
- the residual region (red shaded)
- the MIT-BIH ground-truth symbol + class
- the model's residual, threshold, and decision

Outcome at top of each page:

TP = model caught a true anomaly

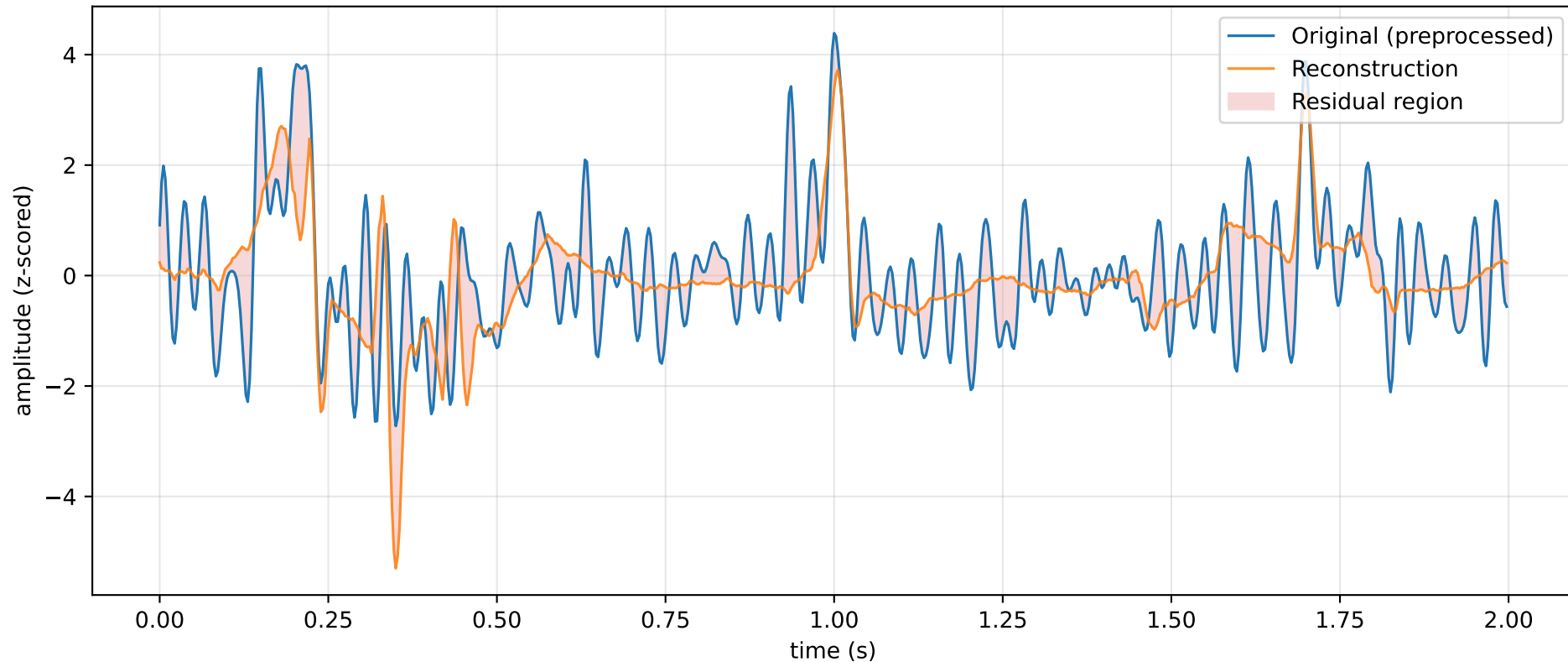
FN = model missed an anomaly

FP = model wrongly flagged a normal beat

TN = model correctly cleared a normal beat

Walk through together; pause on every disagreement between her label and the model's decision.

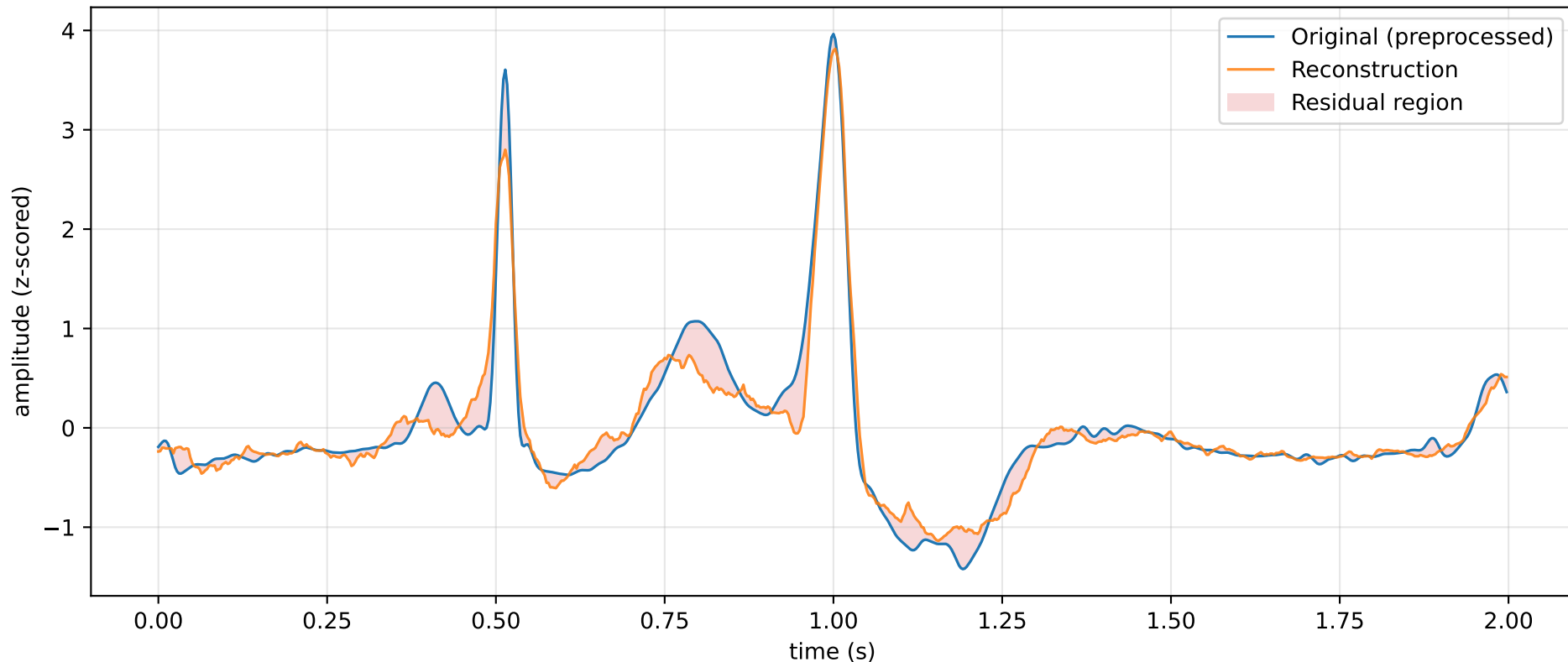
### Beat #1 - TP: ground truth = Unclassifiable (Q)



Source : rec 104, sample 115405  
Beat symbol : Q (Unclassifiable)  
Ground truth : ANOMALY  
Residual : 1.0853  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

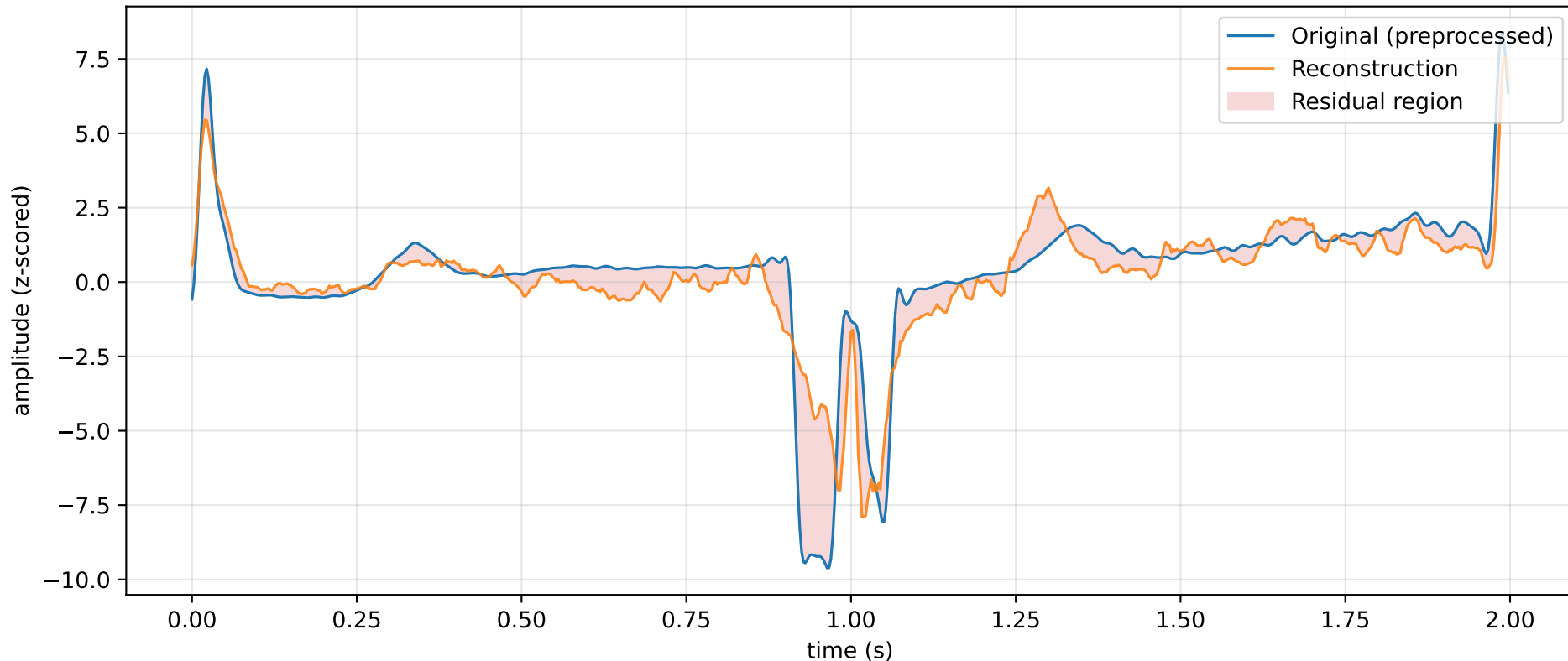
## Beat #2 - FN: ground truth = Premature ventricular contraction (PVC) (V)



Source : rec 106, sample 534246  
Beat symbol : V (Premature ventricular contraction (PVC))  
Ground truth : ANOMALY  
Residual : 0.0427  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

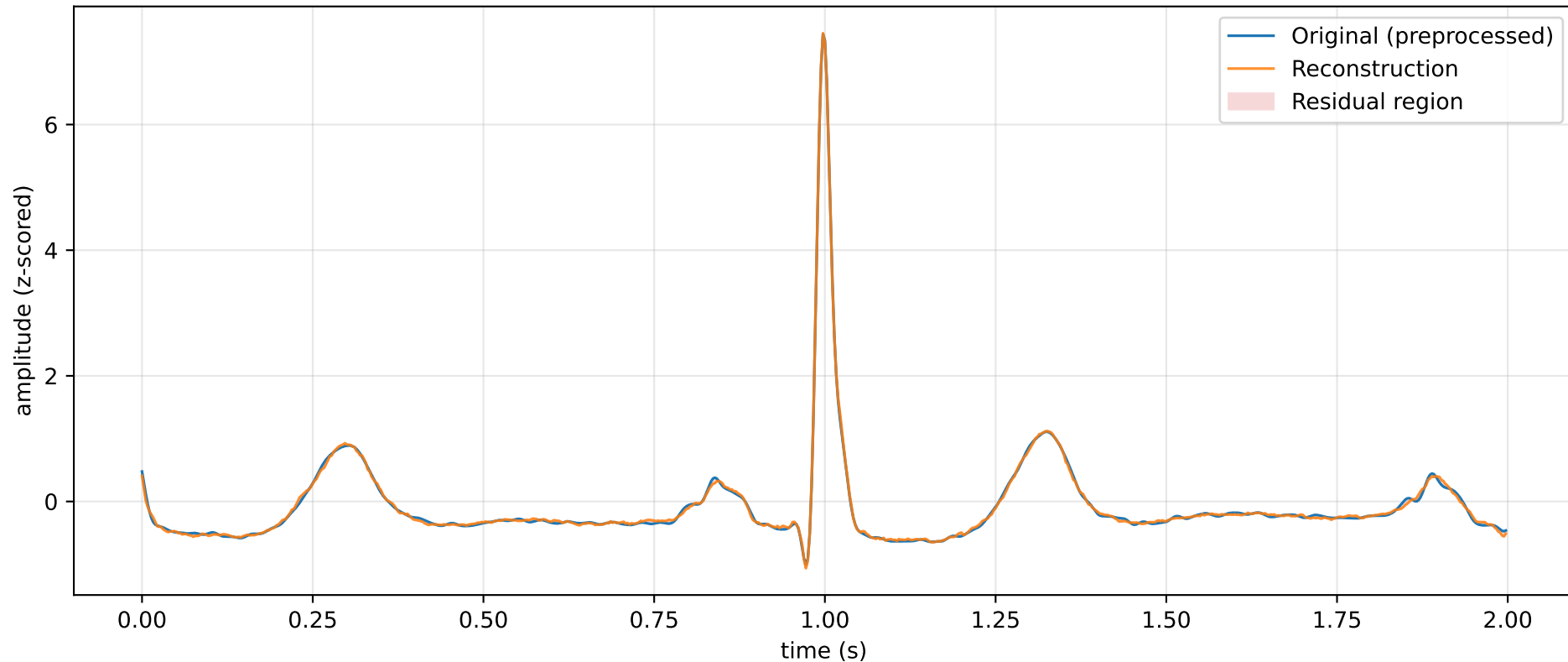
### Beat #3 - TP: ground truth = Unclassifiable (Q)



Source : rec 101, sample 114705  
Beat symbol : Q (Unclassifiable)  
Ground truth : ANOMALY  
Residual : 1.5325  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

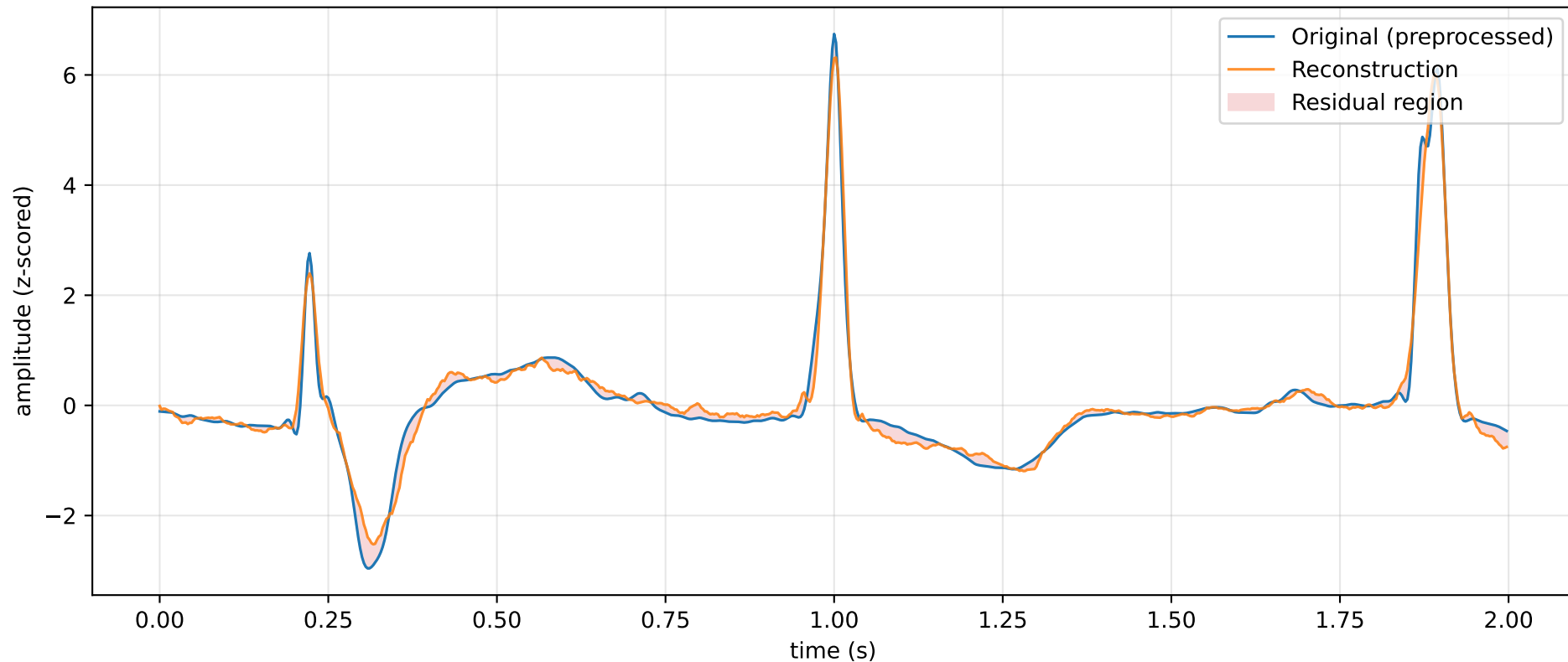
### Beat #4 - TN: ground truth = Normal beat (N)



Source : rec 101, sample 422482  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 0.0009  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

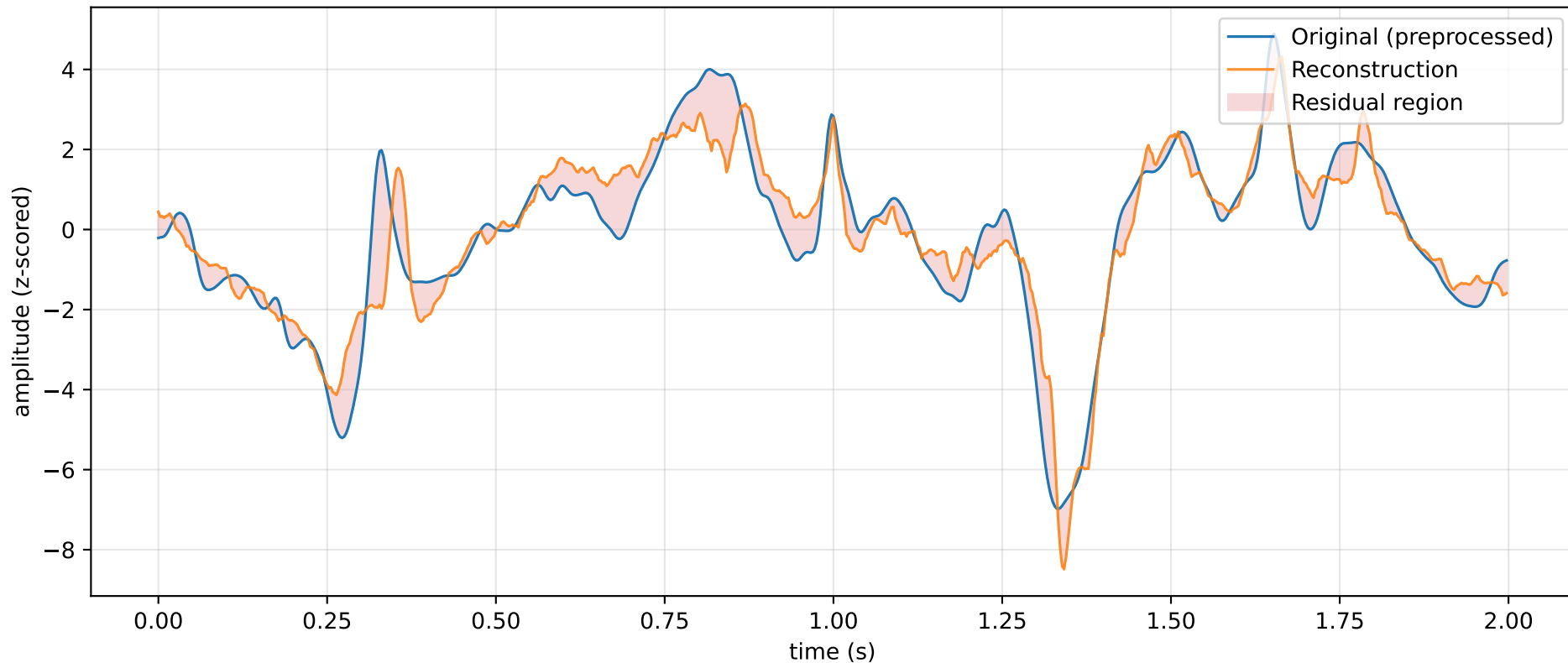
### Beat #5 - FN: ground truth = Fusion of paced and normal (f)



Source : rec 104, sample 597801  
Beat symbol : f (Fusion of paced and normal)  
Ground truth : ANOMALY  
Residual : 0.0421  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

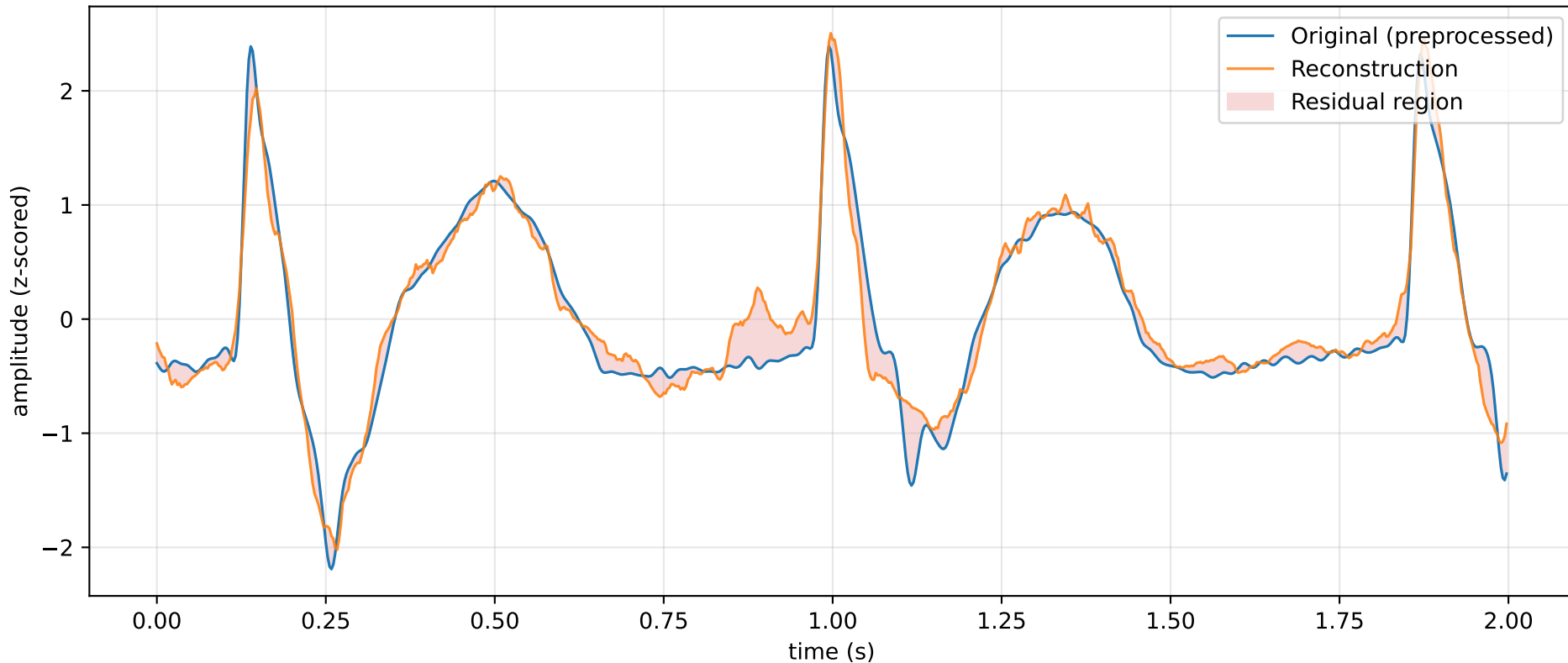
## Beat #6 - FP: ground truth = Normal beat (N)



Source : rec 105, sample 475447  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 0.7276  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

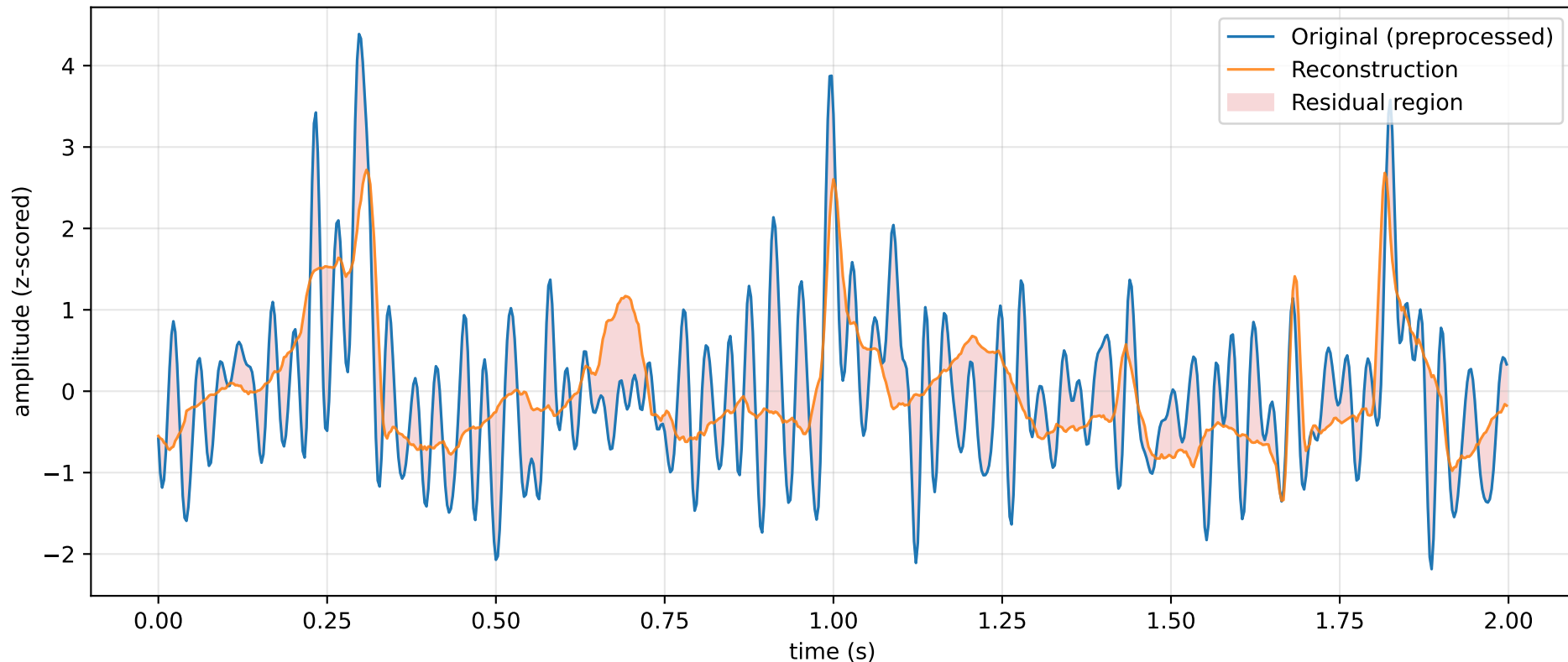
### Beat #7 - FN: ground truth = Paced beat (/)



Source : rec 107, sample 158559  
Beat symbol : / (Paced beat)  
Ground truth : ANOMALY  
Residual : 0.0424  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

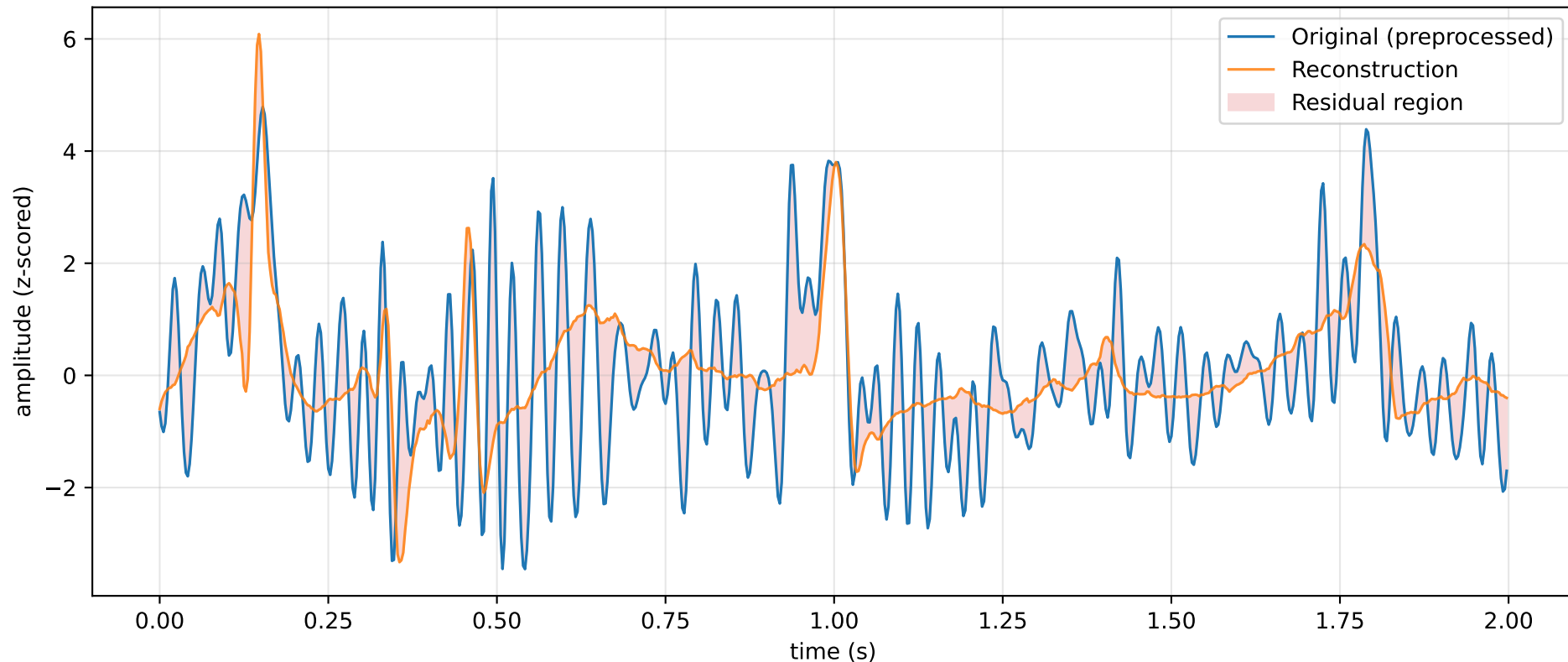
## Beat #8 - TP: ground truth = Unclassifiable (Q)



Source : rec 104, sample 115658  
Beat symbol : Q (Unclassifiable)  
Ground truth : ANOMALY  
Residual : 0.7430  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

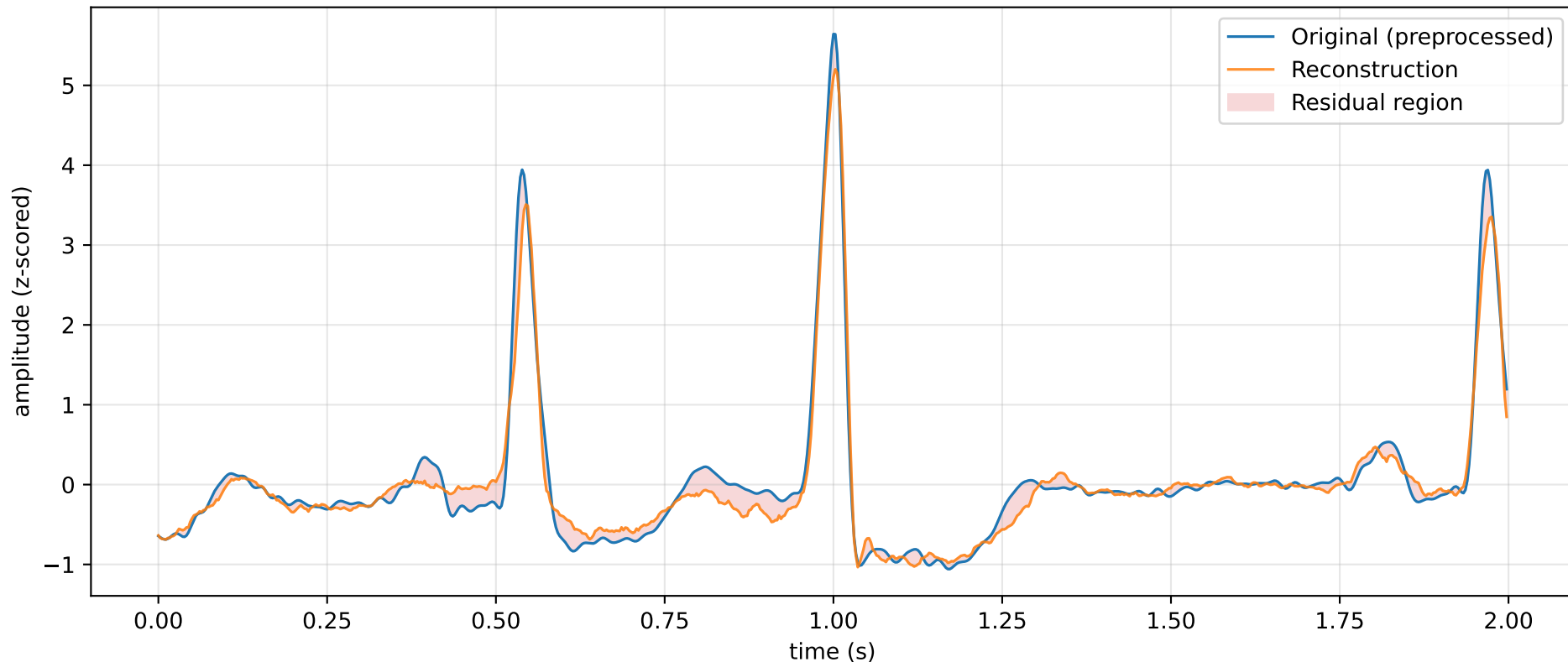
### Beat #9 - TP: ground truth = Unclassifiable (Q)



Source : rec 104, sample 115121  
Beat symbol : Q (Unclassifiable)  
Ground truth : ANOMALY  
Residual : 1.6326  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

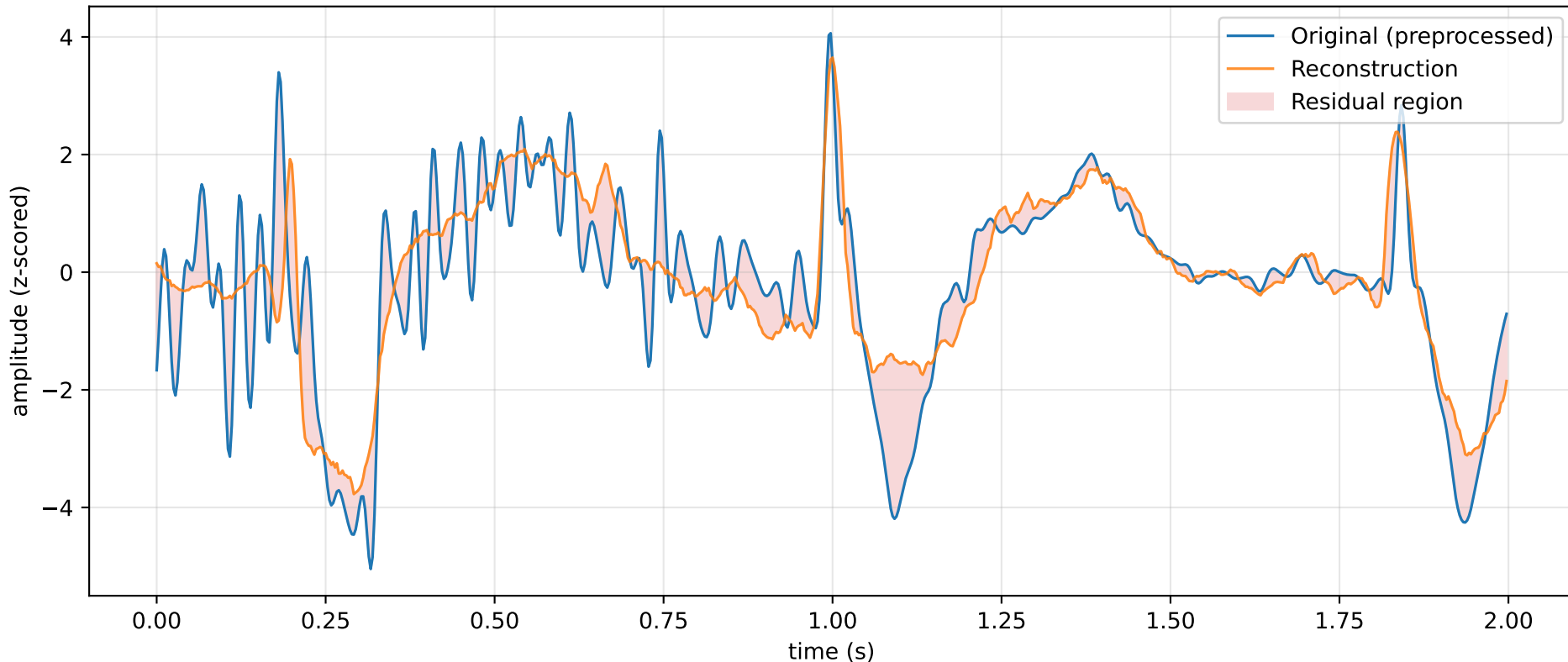
## Beat #10 - FN: ground truth = Premature ventricular contraction (PVC) (V)



Source : rec 105, sample 24842  
Beat symbol : V (Premature ventricular contraction (PVC))  
Ground truth : ANOMALY  
Residual : 0.0425  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

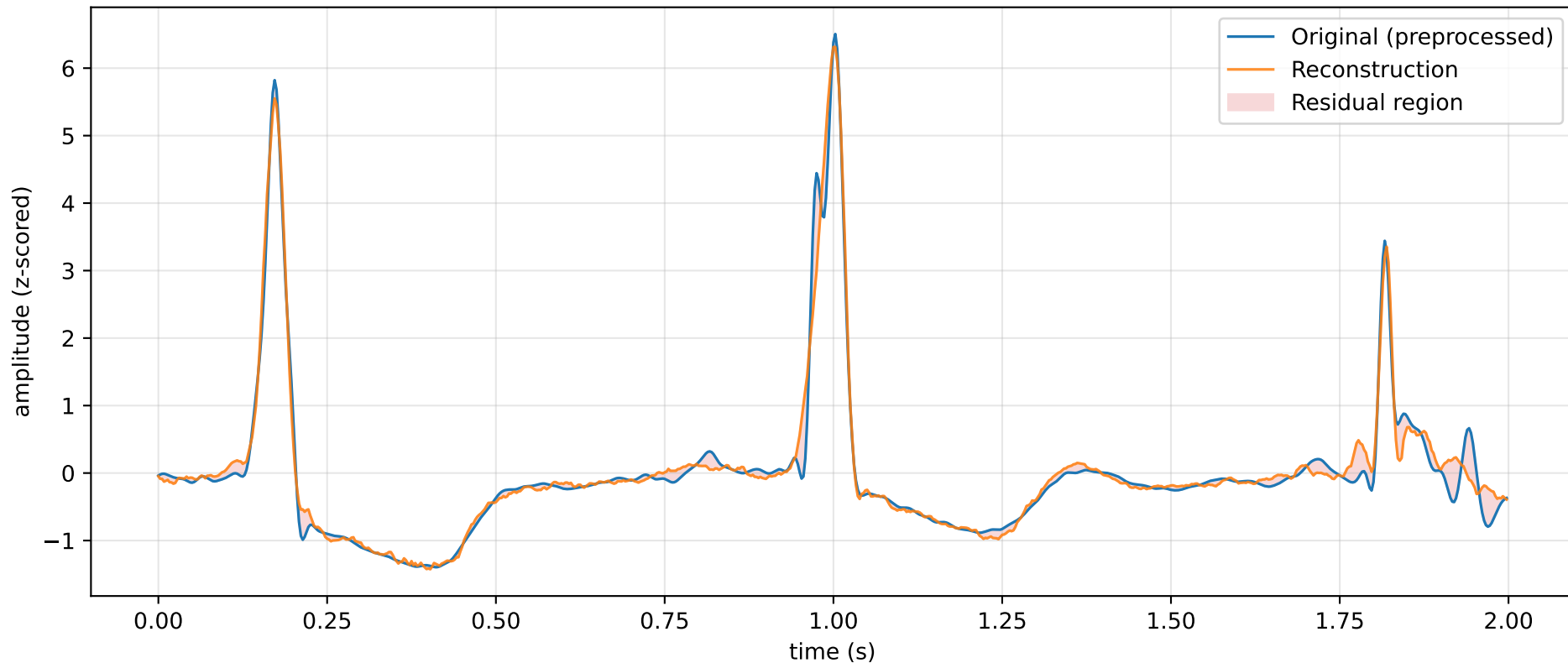
# Beat #11 - TP: ground truth = Paced beat (/)



Source : rec 104, sample 99248  
Beat symbol : / (Paced beat)  
Ground truth : ANOMALY  
Residual : 0.8291  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

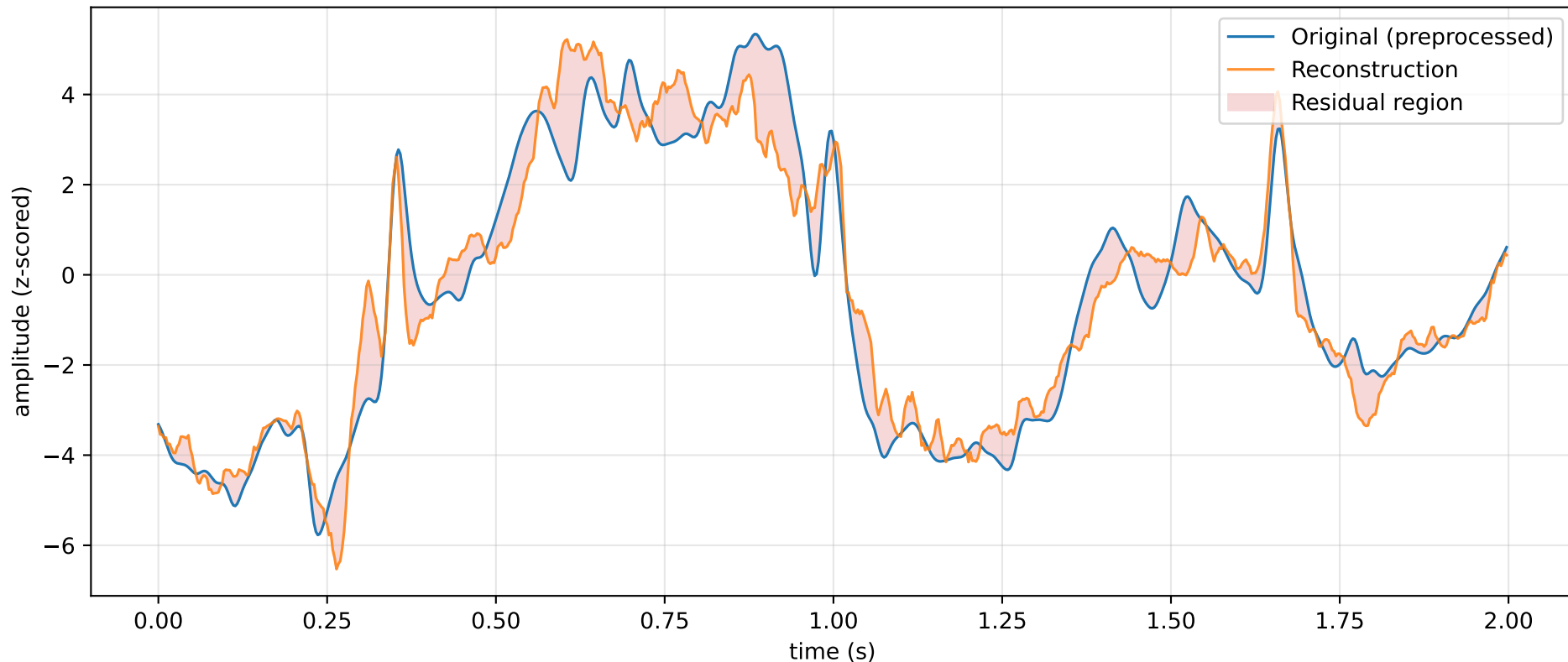
### Beat #12 - FN: ground truth = Fusion of paced and normal (f)



Source : rec 104, sample 299730  
Beat symbol : f (Fusion of paced and normal)  
Ground truth : ANOMALY  
Residual : 0.0421  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

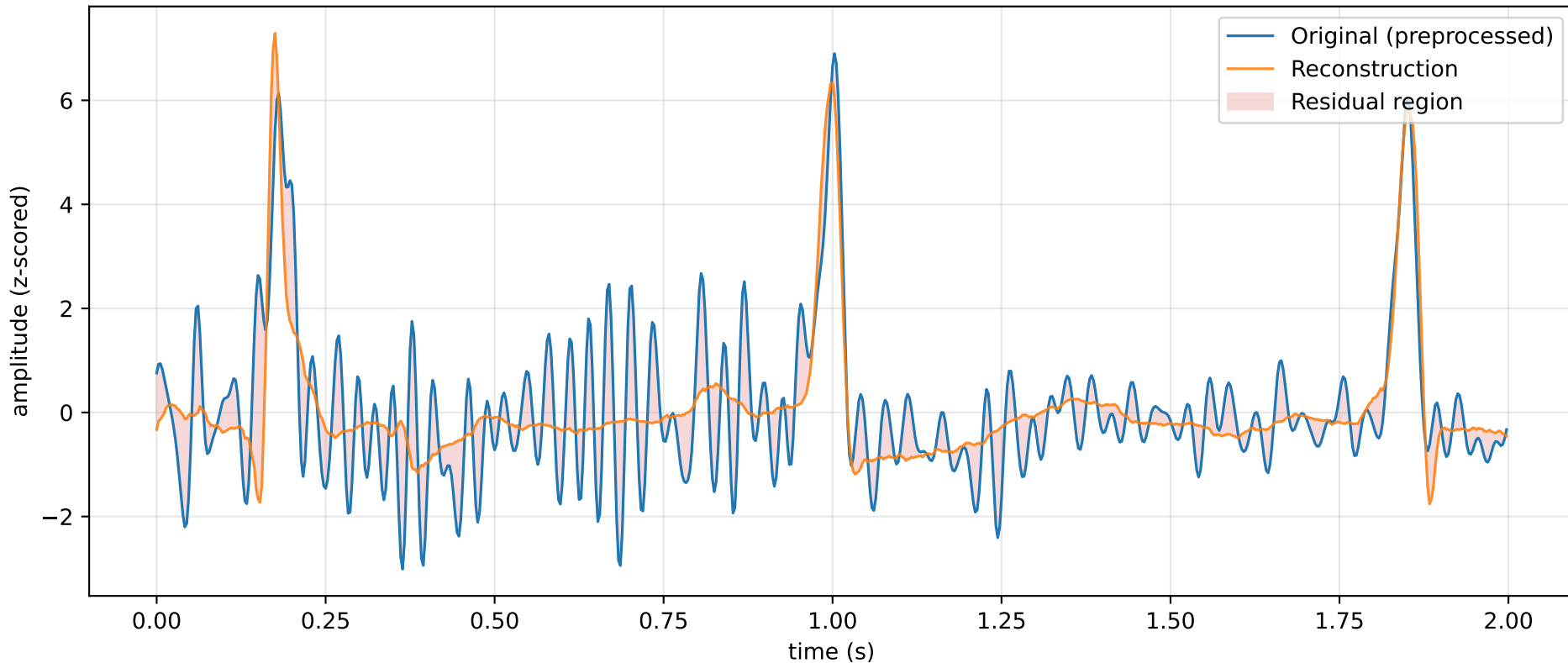
### Beat #13 - FP: ground truth = Normal beat (N)



Source : rec 105, sample 430628  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 0.8842  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

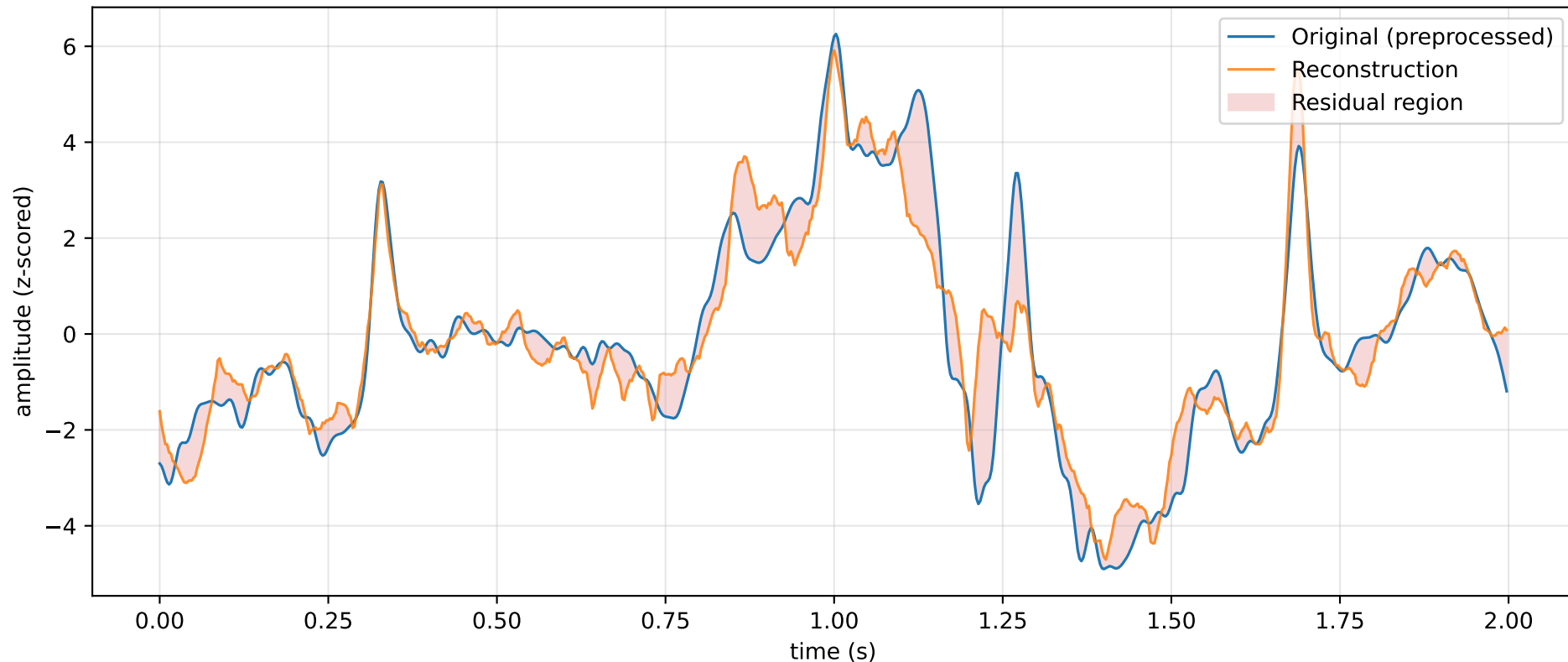
### Beat #14 - TP: ground truth = Unclassifiable (Q)



Source : rec 104, sample 112150  
Beat symbol : Q (Unclassifiable)  
Ground truth : ANOMALY  
Residual : 1.0054  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

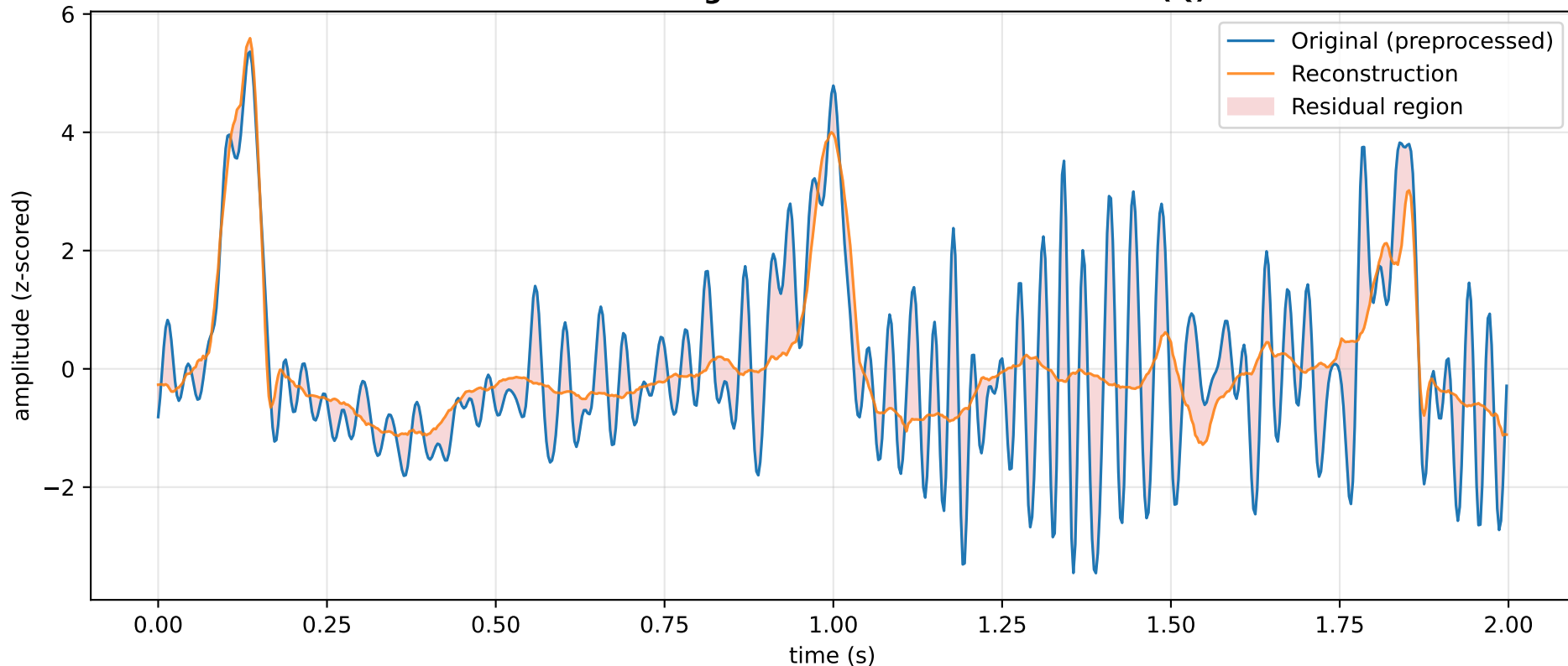
### Beat #15 - FP: ground truth = Normal beat (N)



Source : rec 105, sample 479189  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 0.7792  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

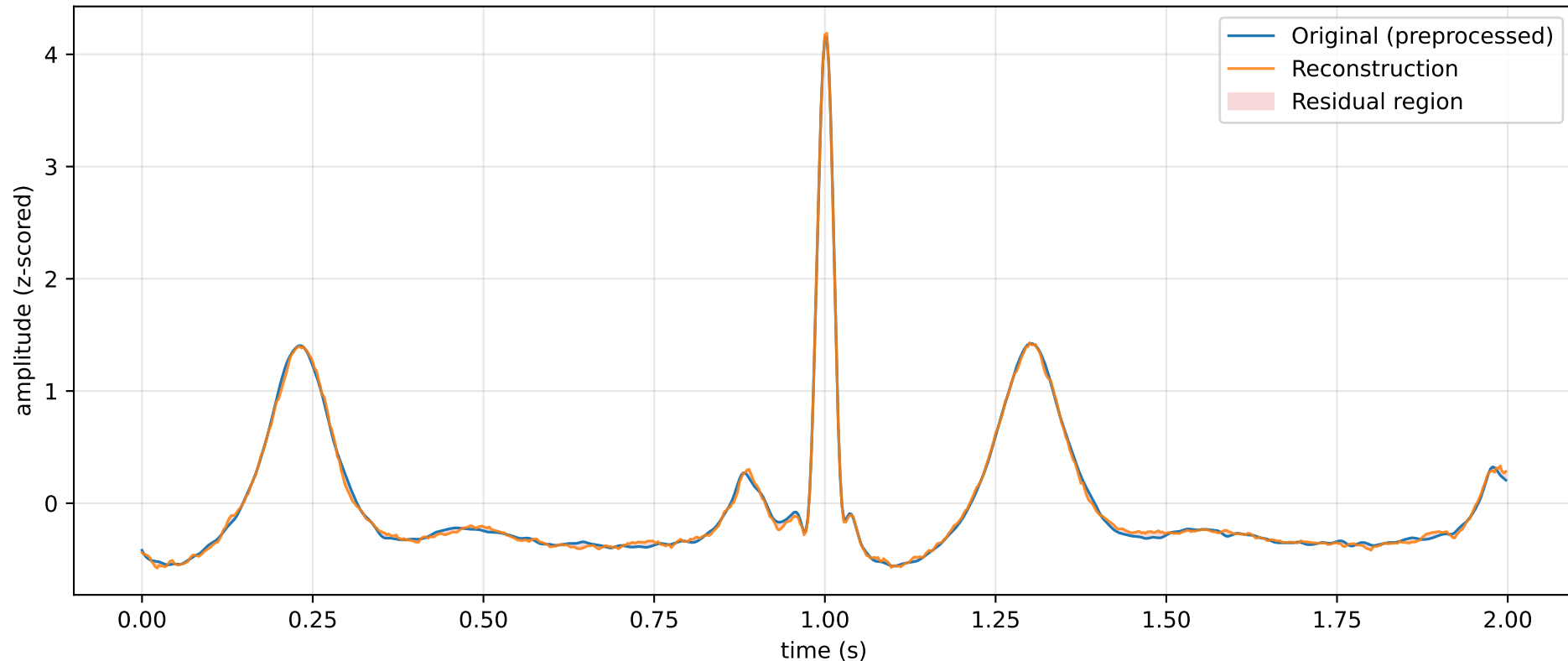
# Beat #16 - TP: ground truth = Unclassifiable (Q)



Source : rec 104, sample 114816  
Beat symbol : Q (Unclassifiable)  
Ground truth : ANOMALY  
Residual : 1.2485  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

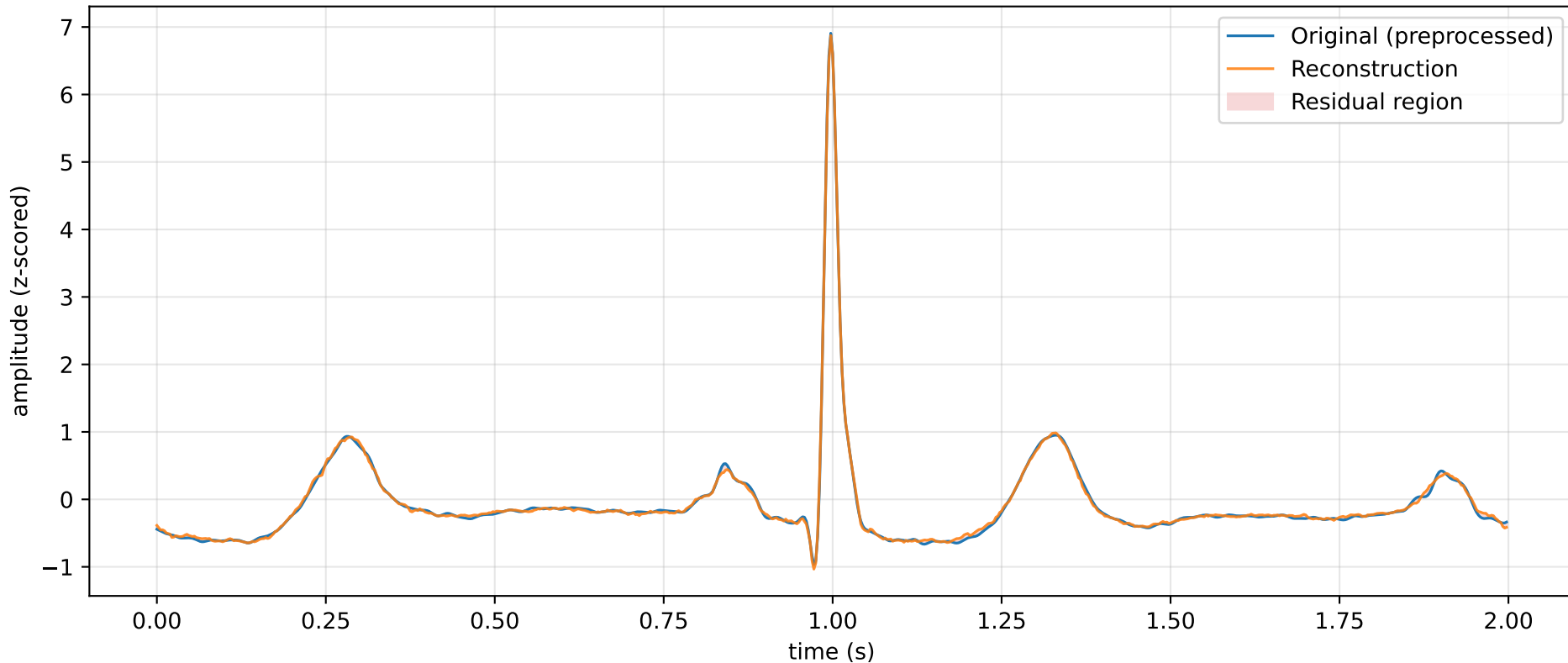
### Beat #17 - TN: ground truth = Normal beat (N)



Source : rec 106, sample 109668  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 0.0008  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

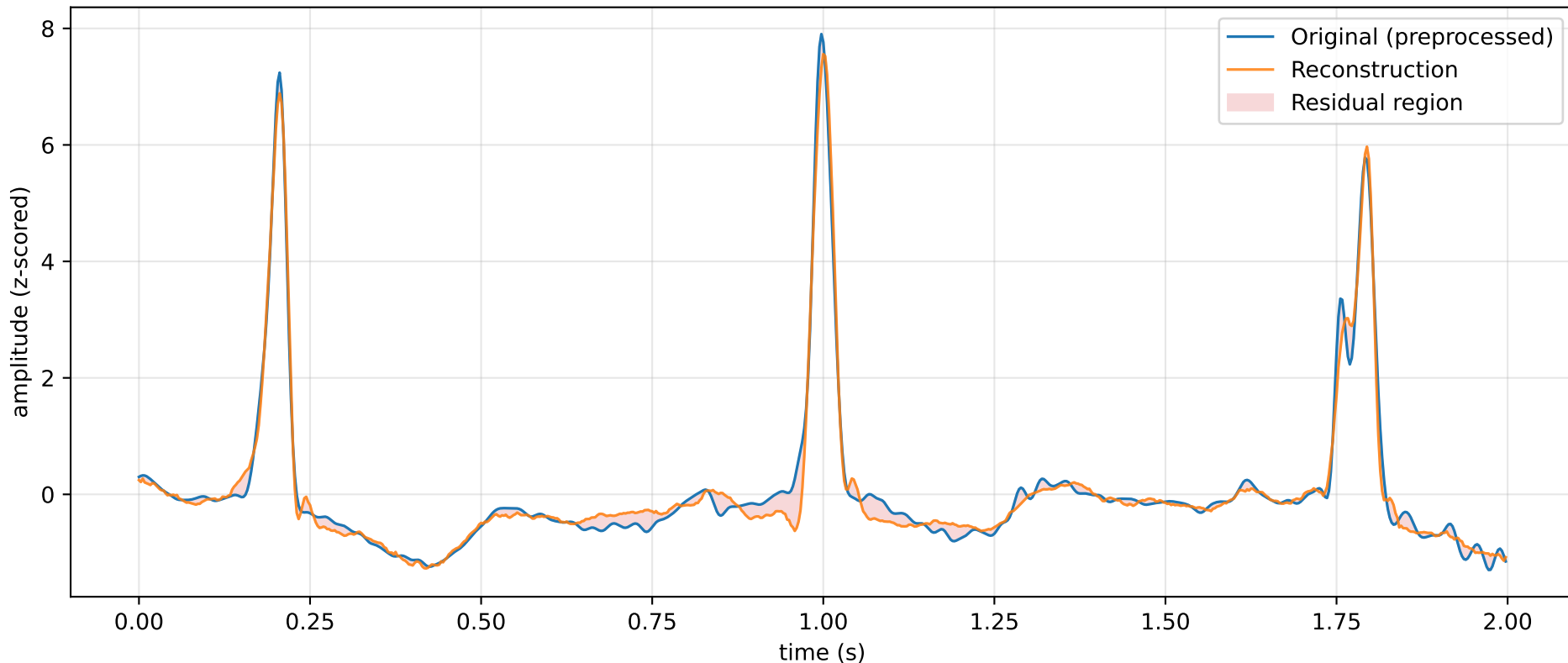
### Beat #18 - TN: ground truth = Normal beat (N)



Source : rec 101, sample 254123  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 0.0009  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

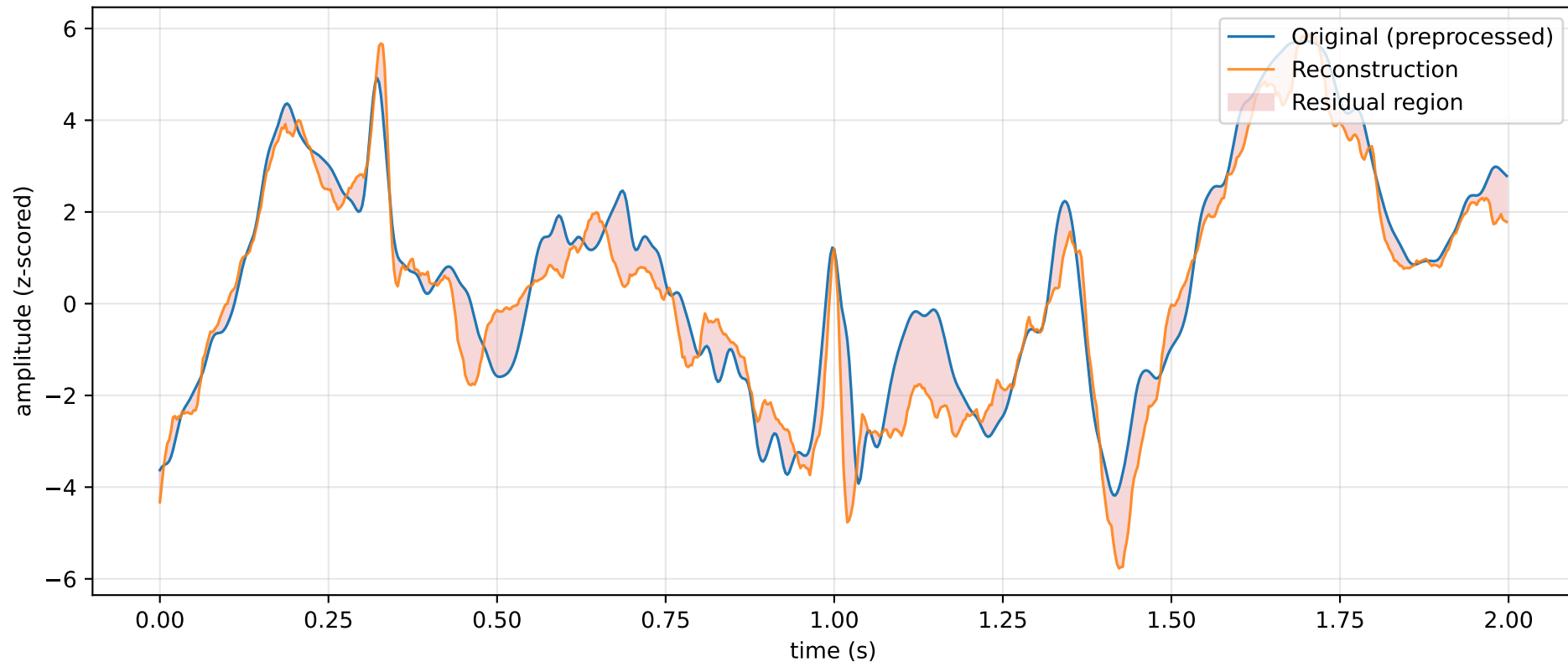
# Beat #19 - FN: ground truth = Fusion of paced and normal (f)



Source : rec 104, sample 184873  
Beat symbol : f (Fusion of paced and normal)  
Ground truth : ANOMALY  
Residual : 0.0430  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

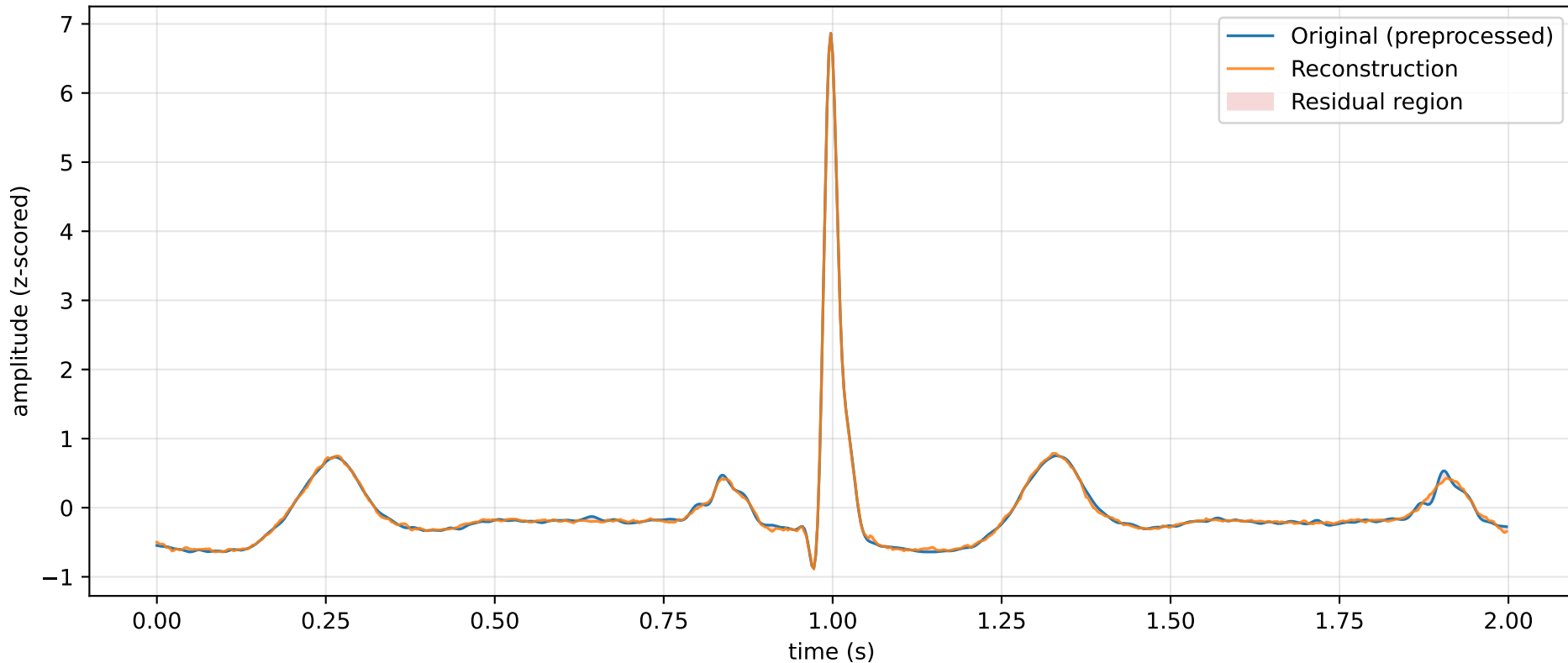
### Beat #20 - FP: ground truth = Normal beat (N)



Source : rec 105, sample 433506  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 0.7122  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

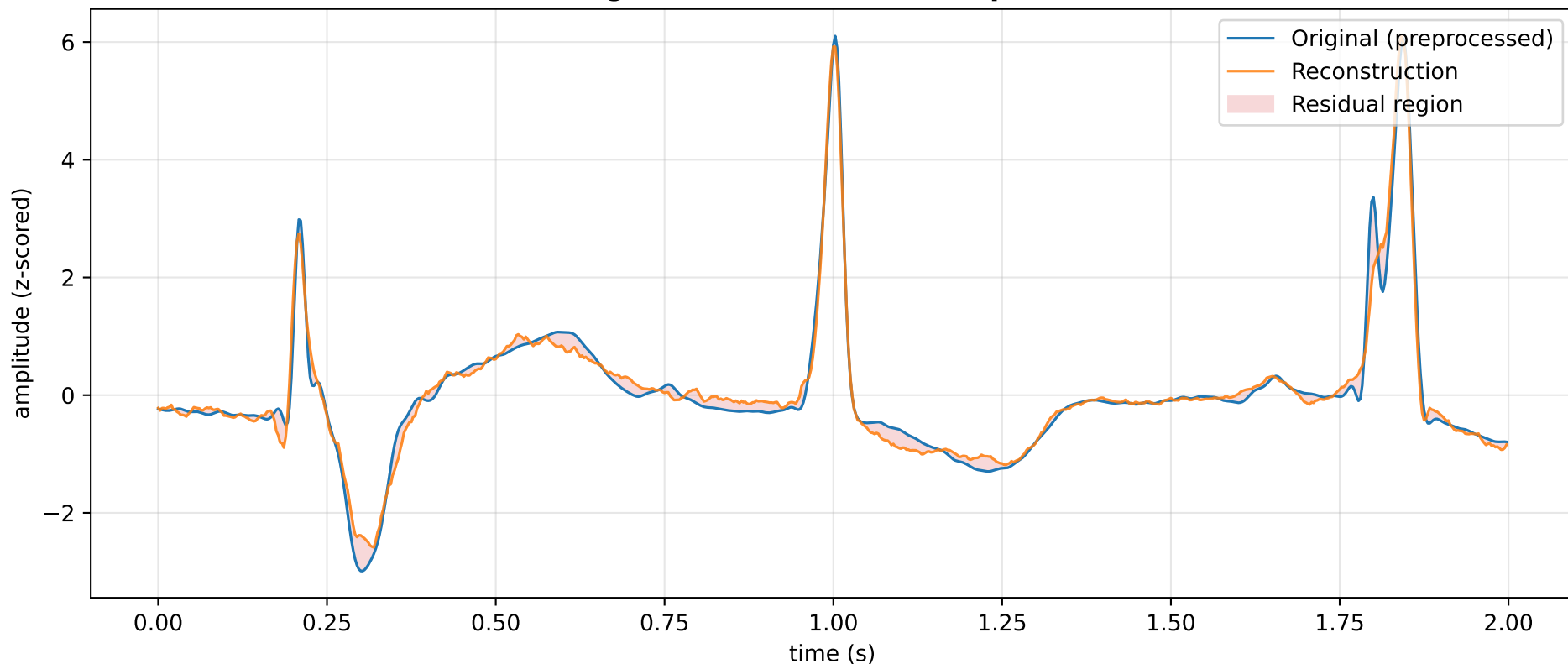
### Beat #21 - TN: ground truth = Normal beat (N)



Source : rec 101, sample 630244  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 0.0009  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

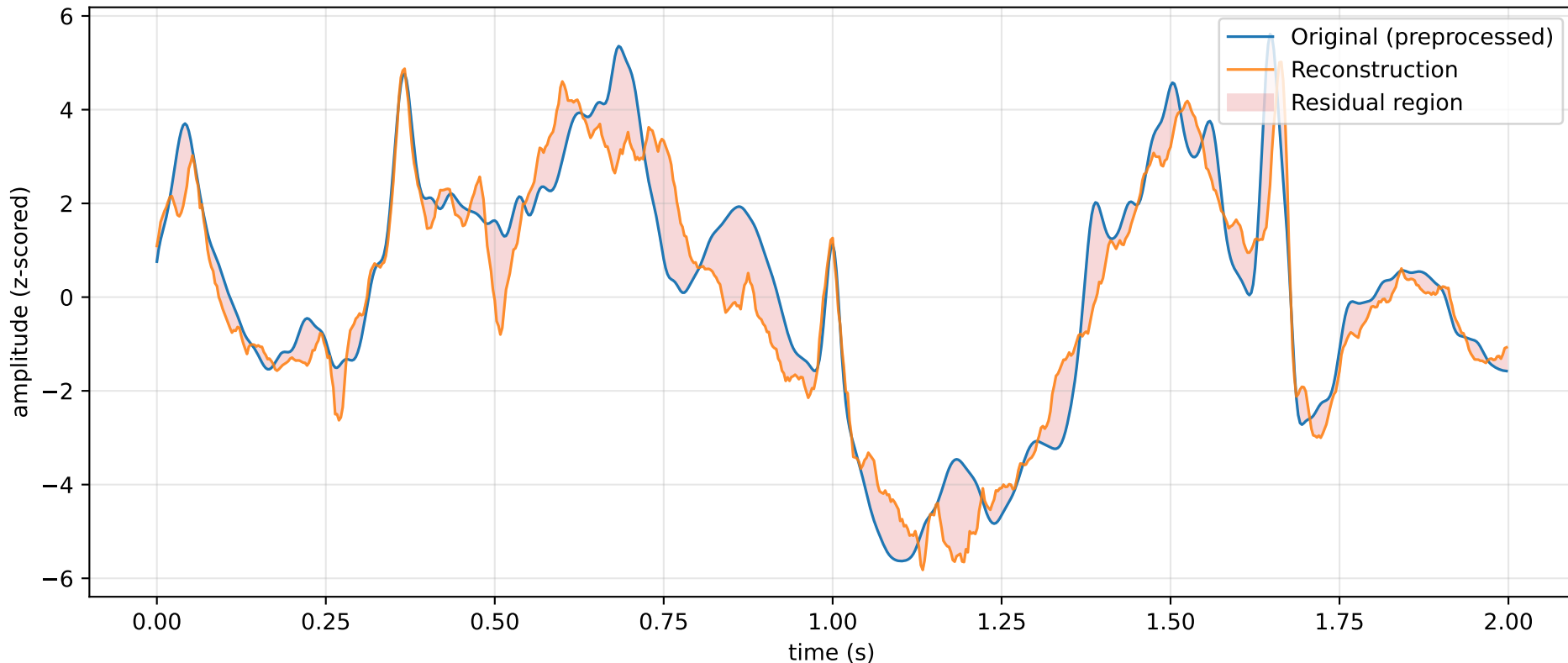
## Beat #22 - FN: ground truth = Fusion of paced and normal (f)



Source : rec 104, sample 592849  
Beat symbol : f (Fusion of paced and normal)  
Ground truth : ANOMALY  
Residual : 0.0420  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

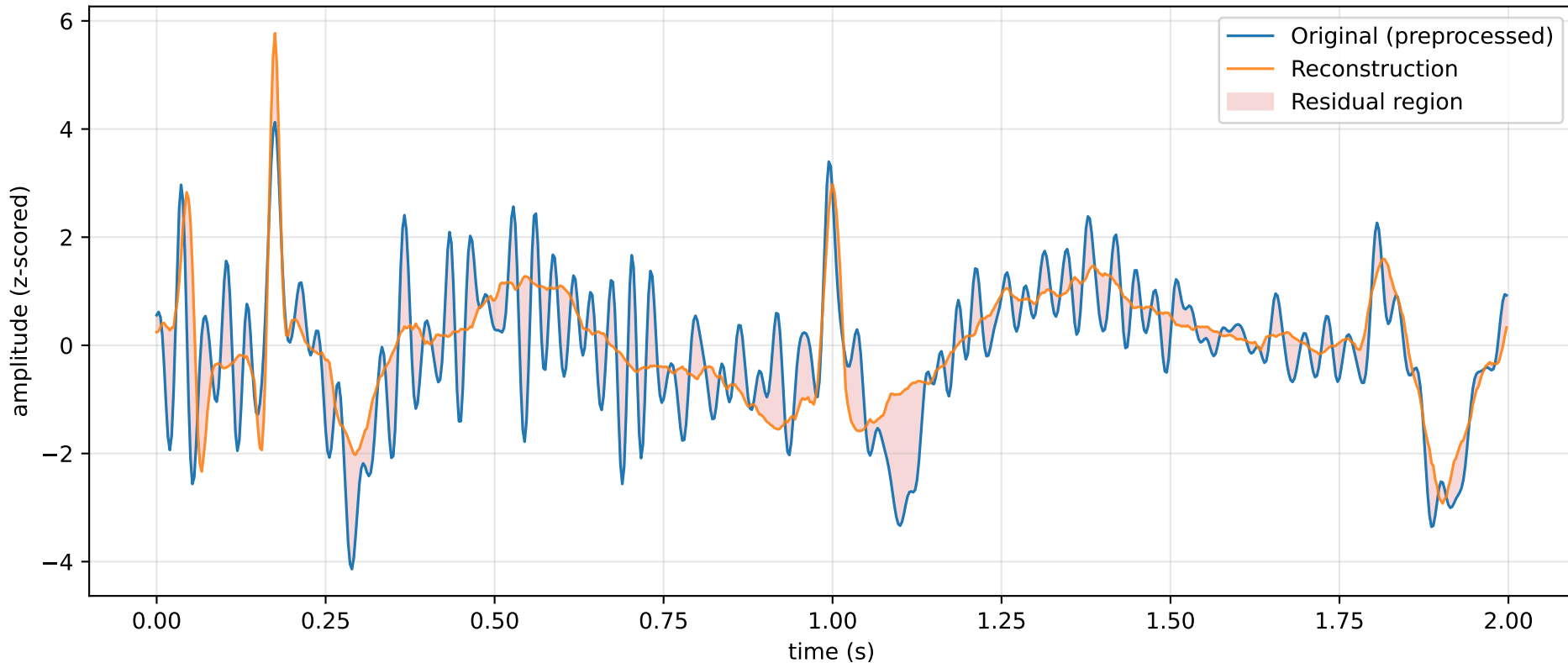
### Beat #23 - FP: ground truth = Normal beat (N)



Source : rec 105, sample 441811  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 0.8911  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

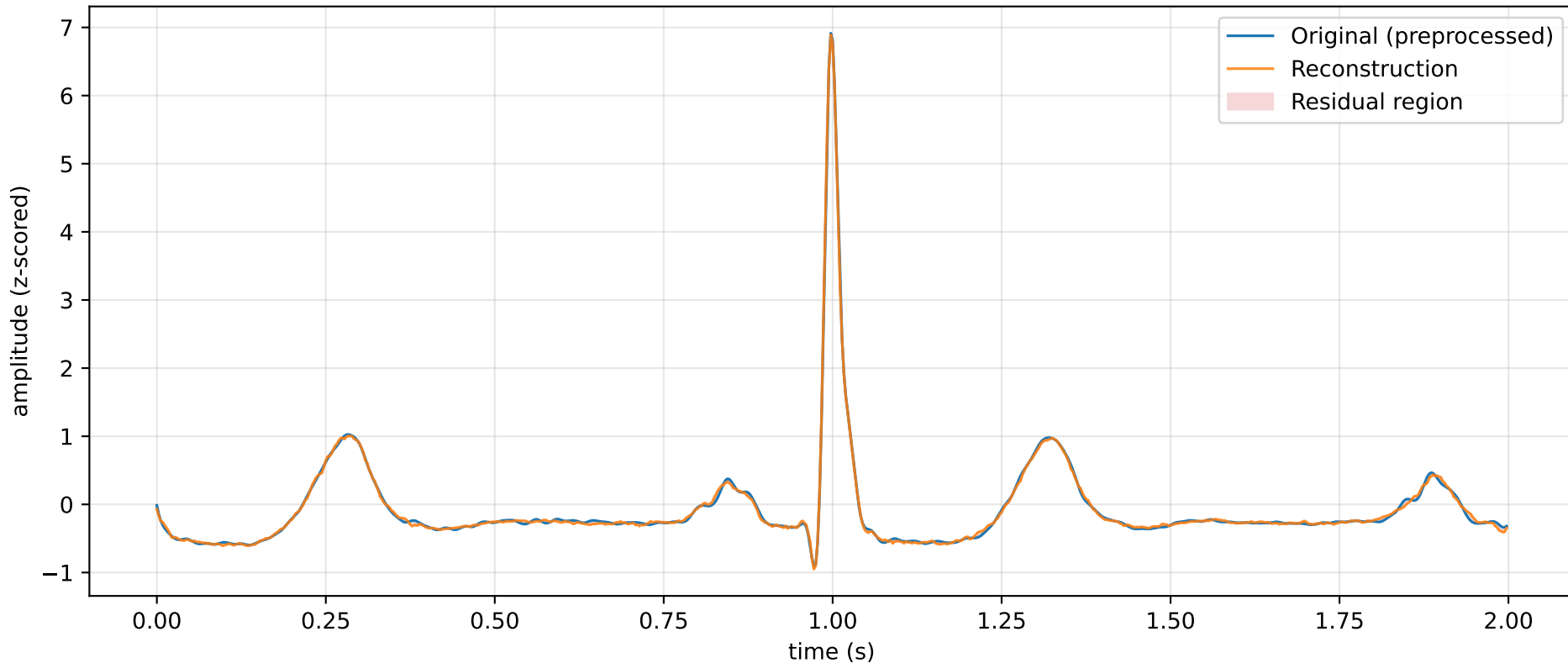
## Beat #24 - TP: ground truth = Paced beat (/)



Source : rec 104, sample 48232  
Beat symbol : / (Paced beat)  
Ground truth : ANOMALY  
Residual : 0.8561  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

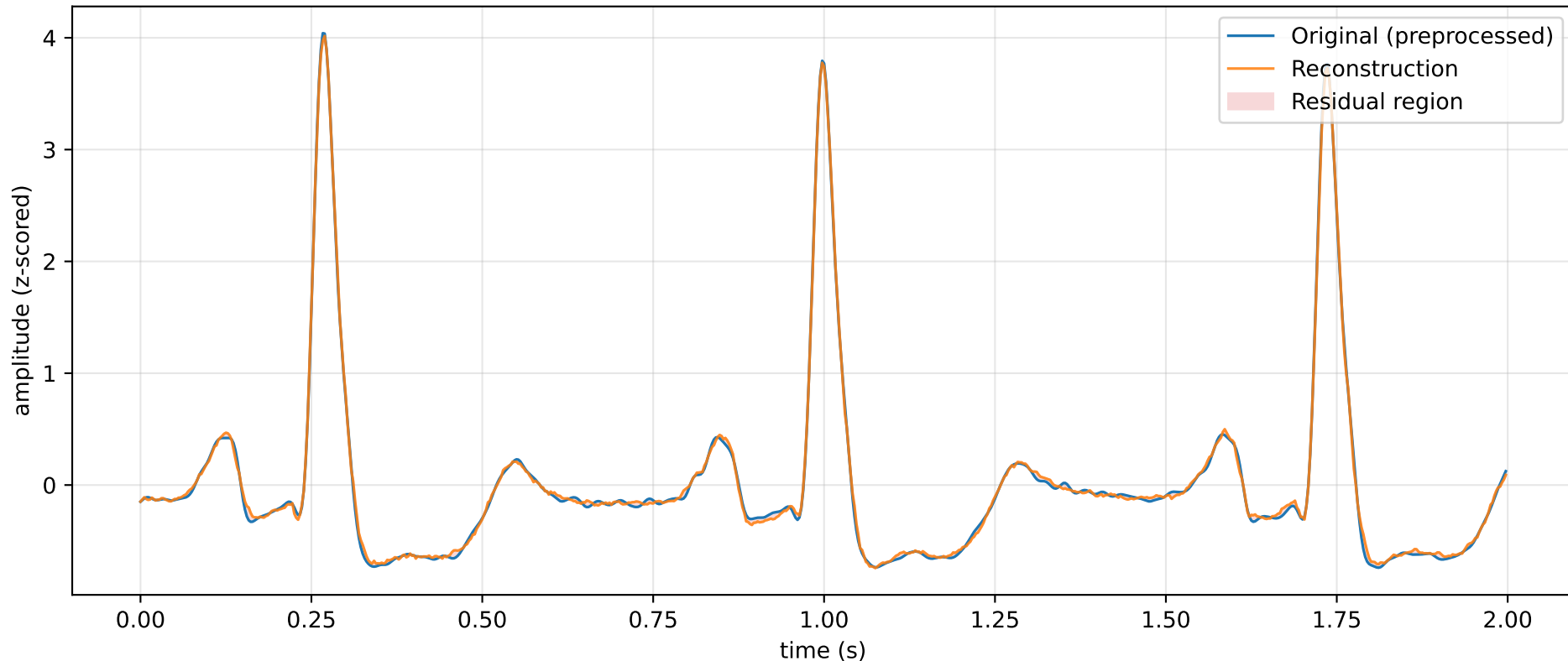
### Beat #25 - TN: ground truth = Normal beat (N)



Source : rec 101, sample 385980  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 0.0008  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

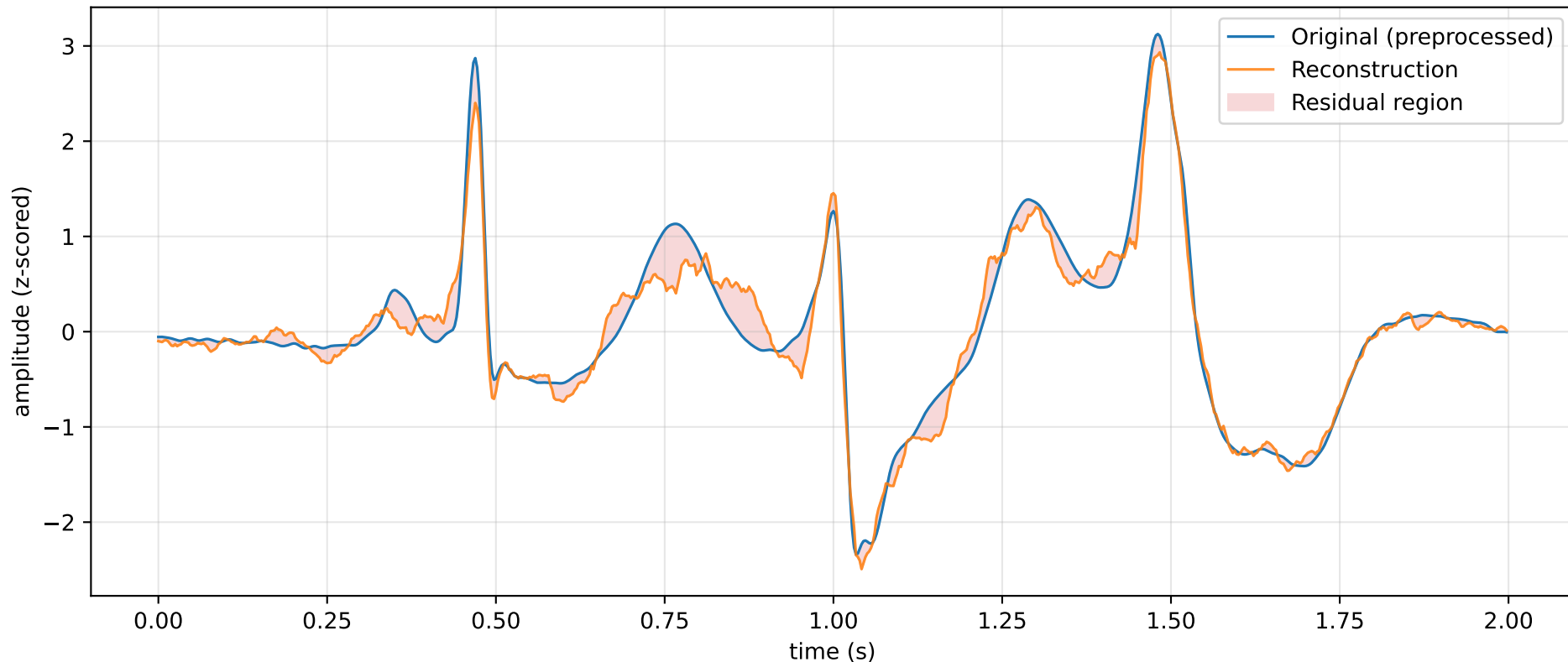
## Beat #26 - TN: ground truth = Normal beat (N)



Source : rec 105, sample 131954  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 0.0009  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

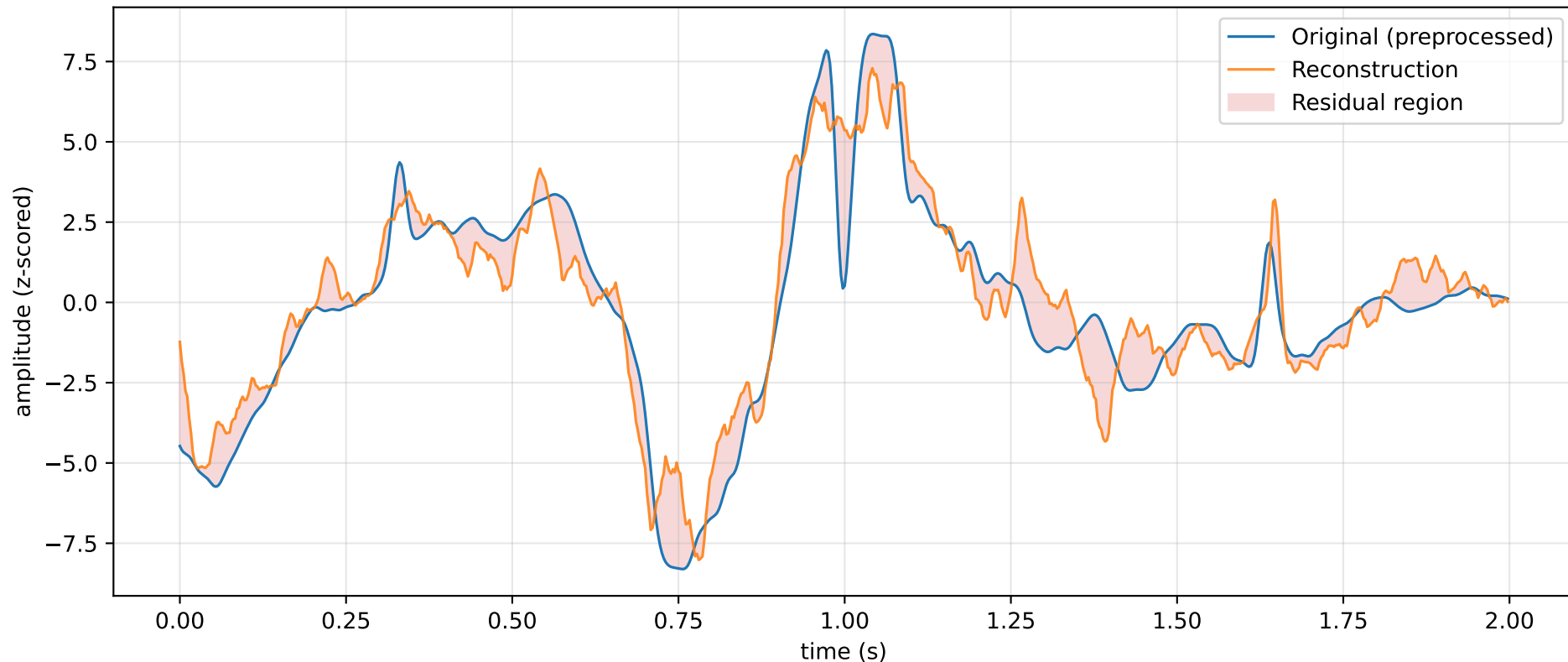
## Beat #27 - FN: ground truth = Premature ventricular contraction (PVC) (V)



Source : rec 106, sample 611355  
Beat symbol : V (Premature ventricular contraction (PVC))  
Ground truth : ANOMALY  
Residual : 0.0428  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

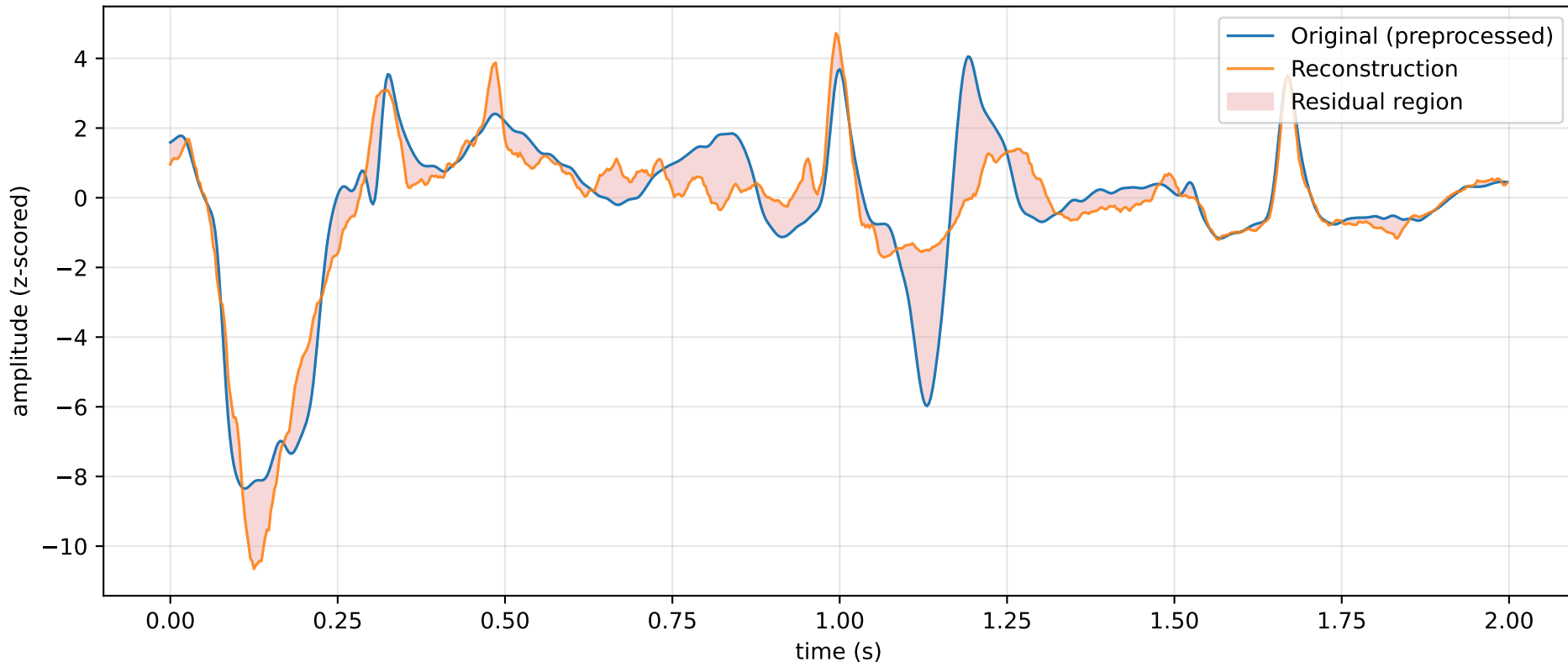
## Beat #28 - TP: ground truth = Unclassifiable (Q)



Source : rec 105, sample 444182  
Beat symbol : Q (Unclassifiable)  
Ground truth : ANOMALY  
Residual : 1.6444  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

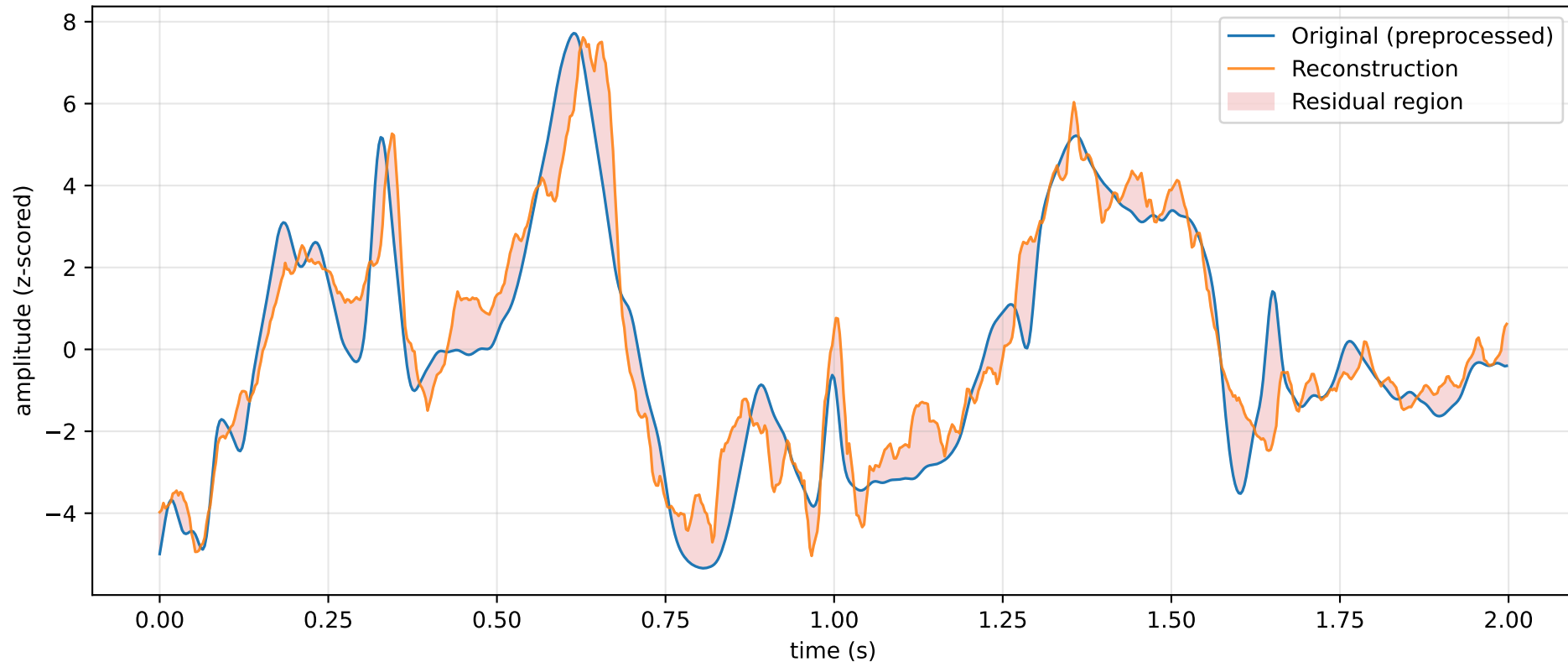
## Beat #29 - FP: ground truth = Normal beat (N)



Source : rec 105, sample 457427  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 1.1841  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

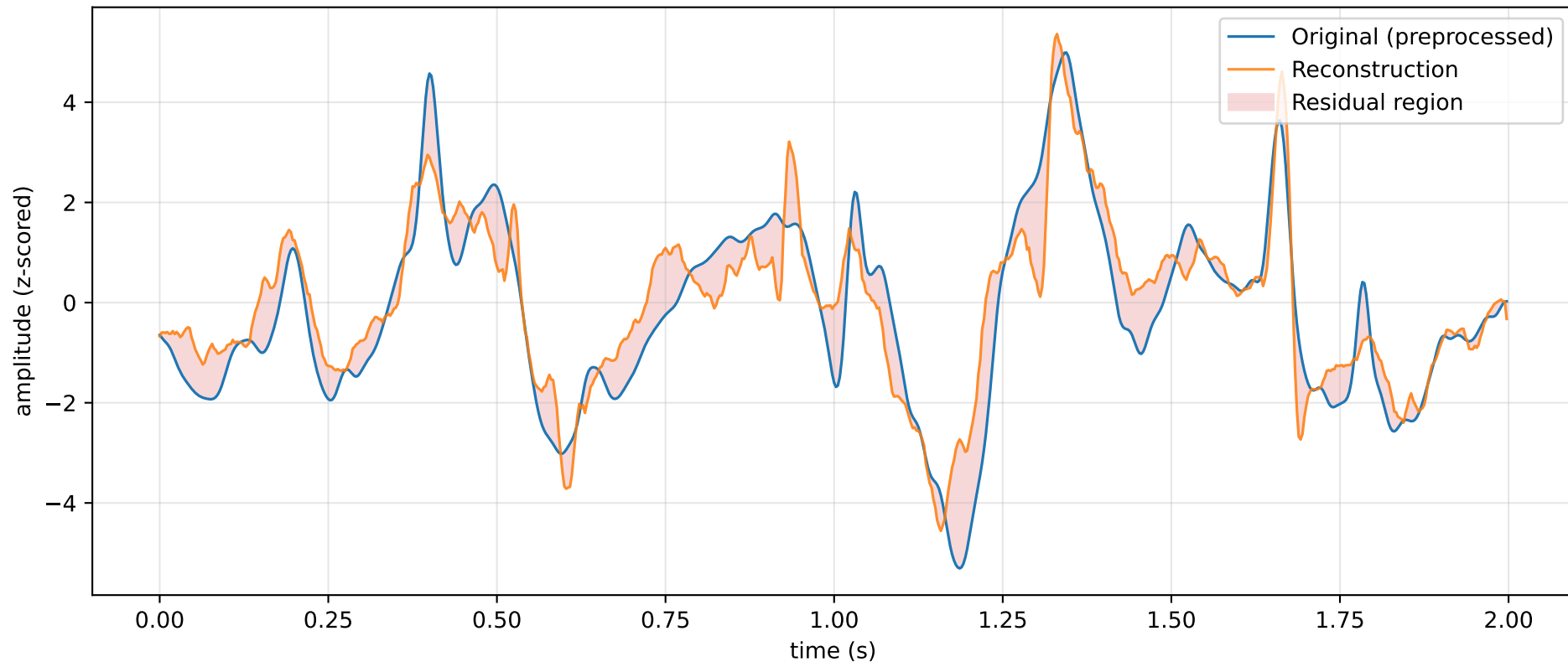
### Beat #30 - FP: ground truth = Normal beat (N)



Source : rec 105, sample 477791  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 1.1486  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

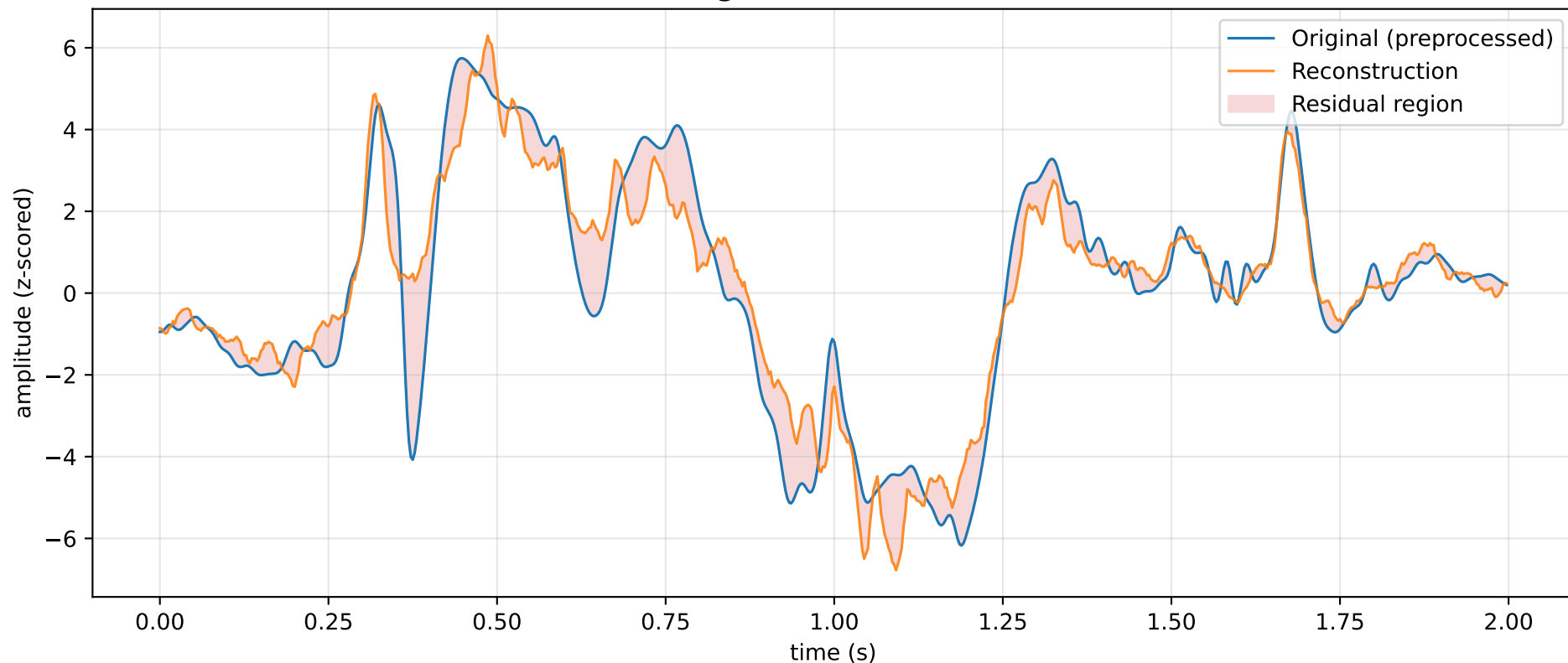
### Beat #31 - FP: ground truth = Normal beat (N)



Source : rec 105, sample 476122  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 0.7322  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

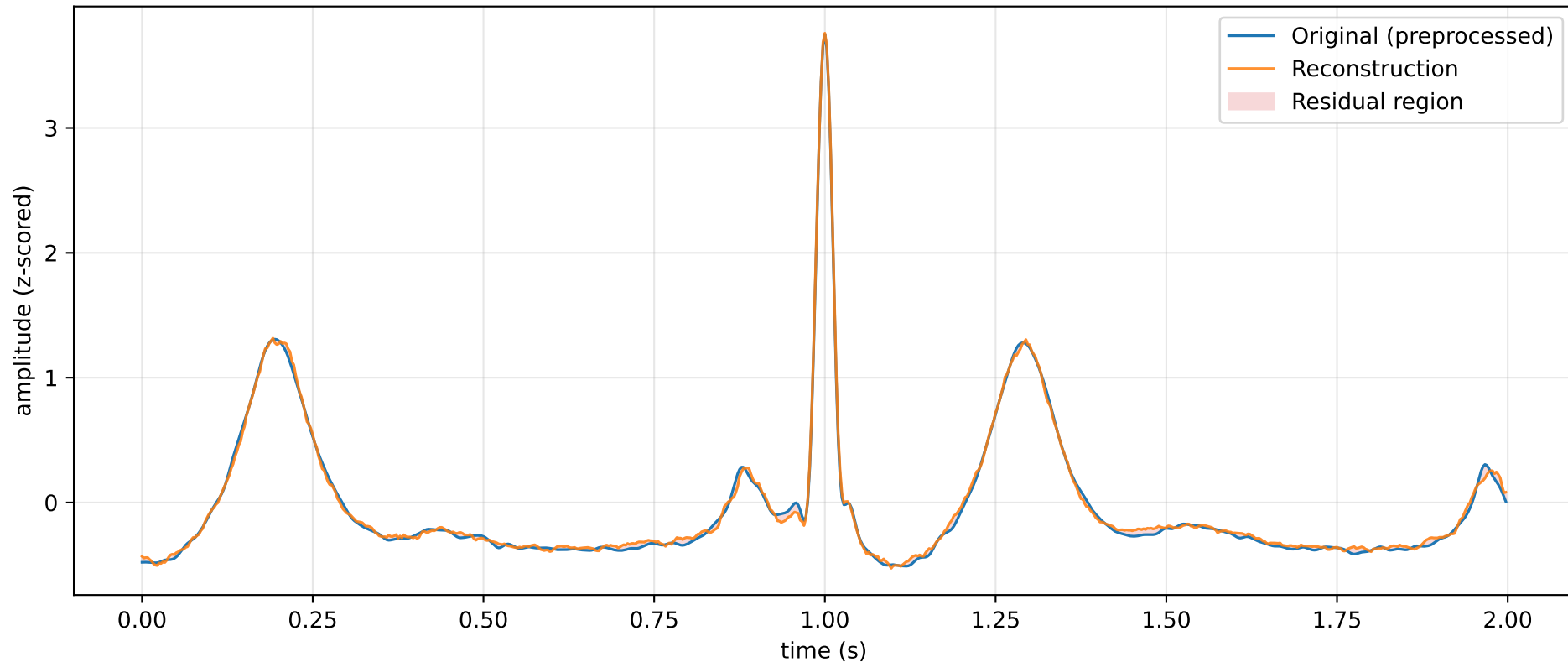
### Beat #32 - FP: ground truth = Normal beat (N)



Source : rec 105, sample 435895  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 1.0606  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

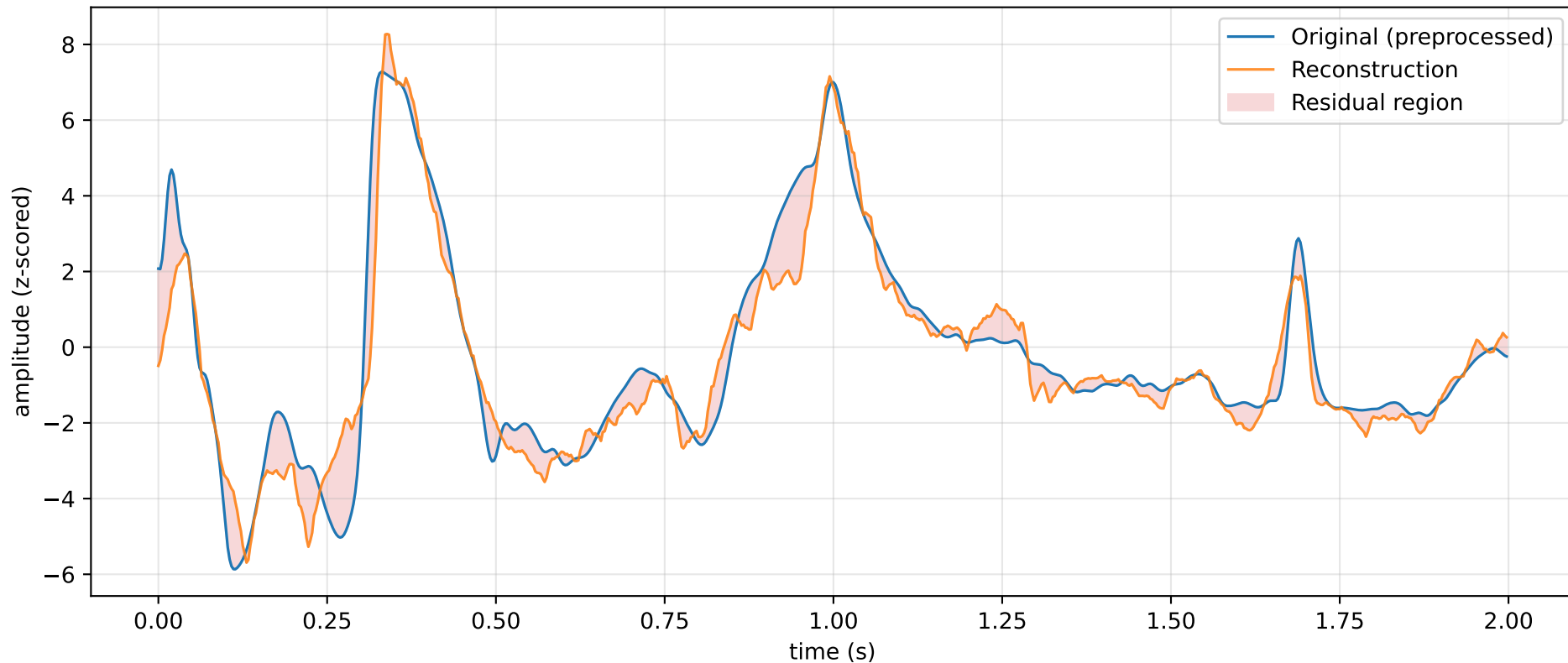
### Beat #33 - TN: ground truth = Normal beat (N)



Source : rec 106, sample 105898  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 0.0009  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

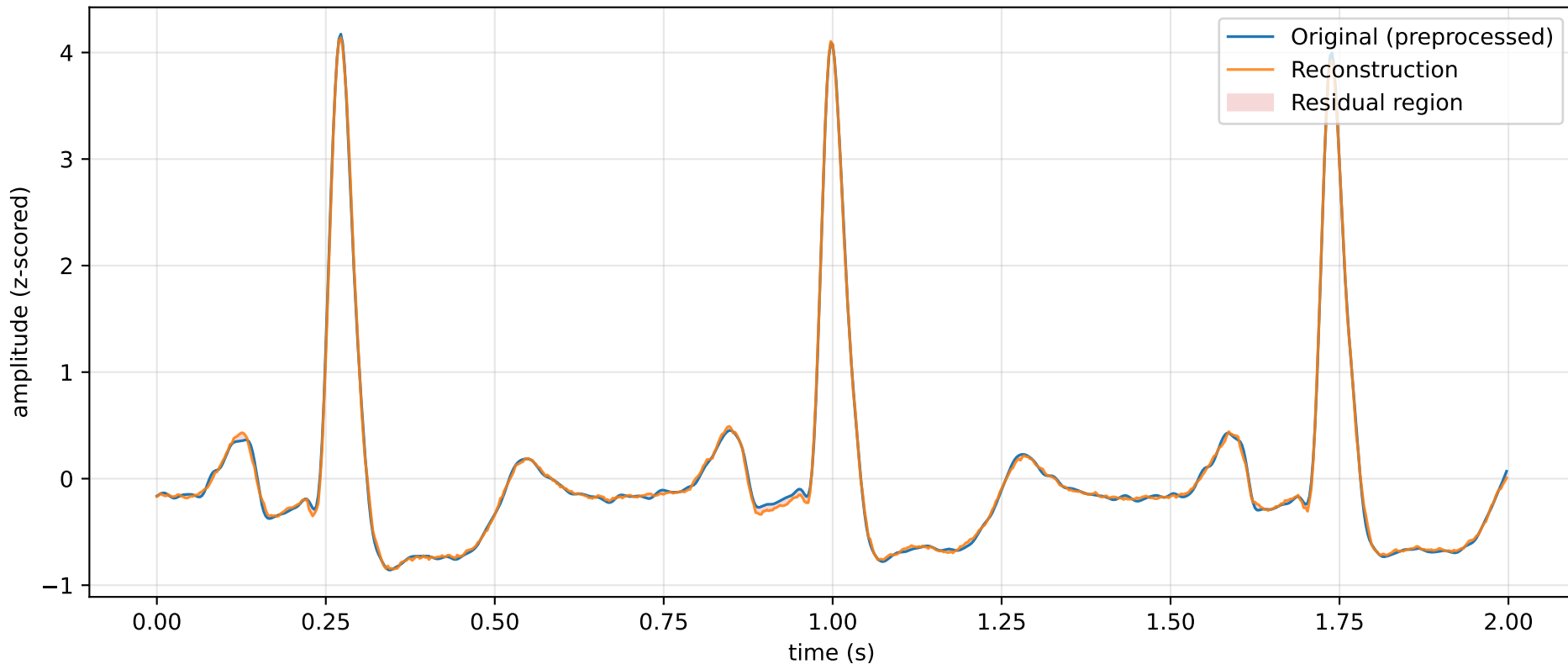
### Beat #34 - TP: ground truth = Unclassifiable (Q)



Source : rec 105, sample 595482  
Beat symbol : Q (Unclassifiable)  
Ground truth : ANOMALY  
Residual : 0.8405  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

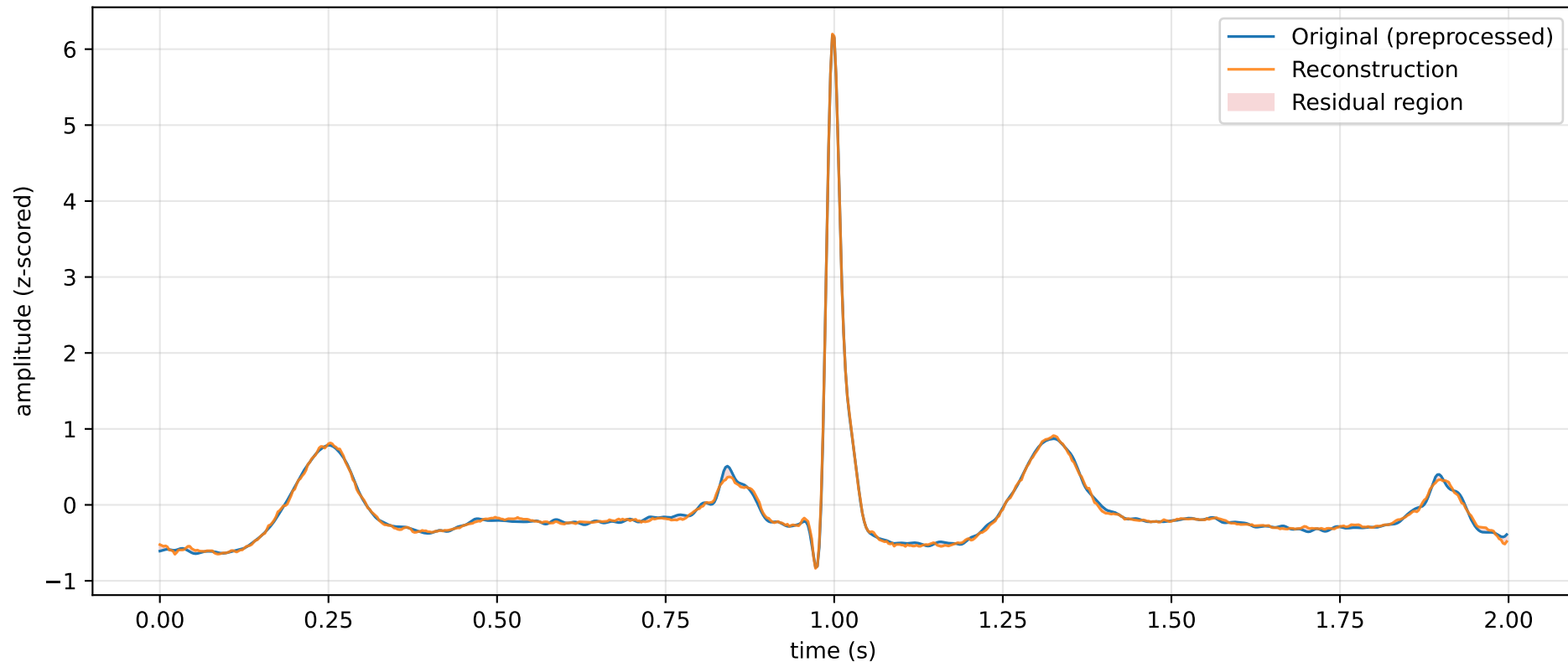
### Beat #35 - TN: ground truth = Normal beat (N)



Source : rec 105, sample 153530  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 0.0008  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

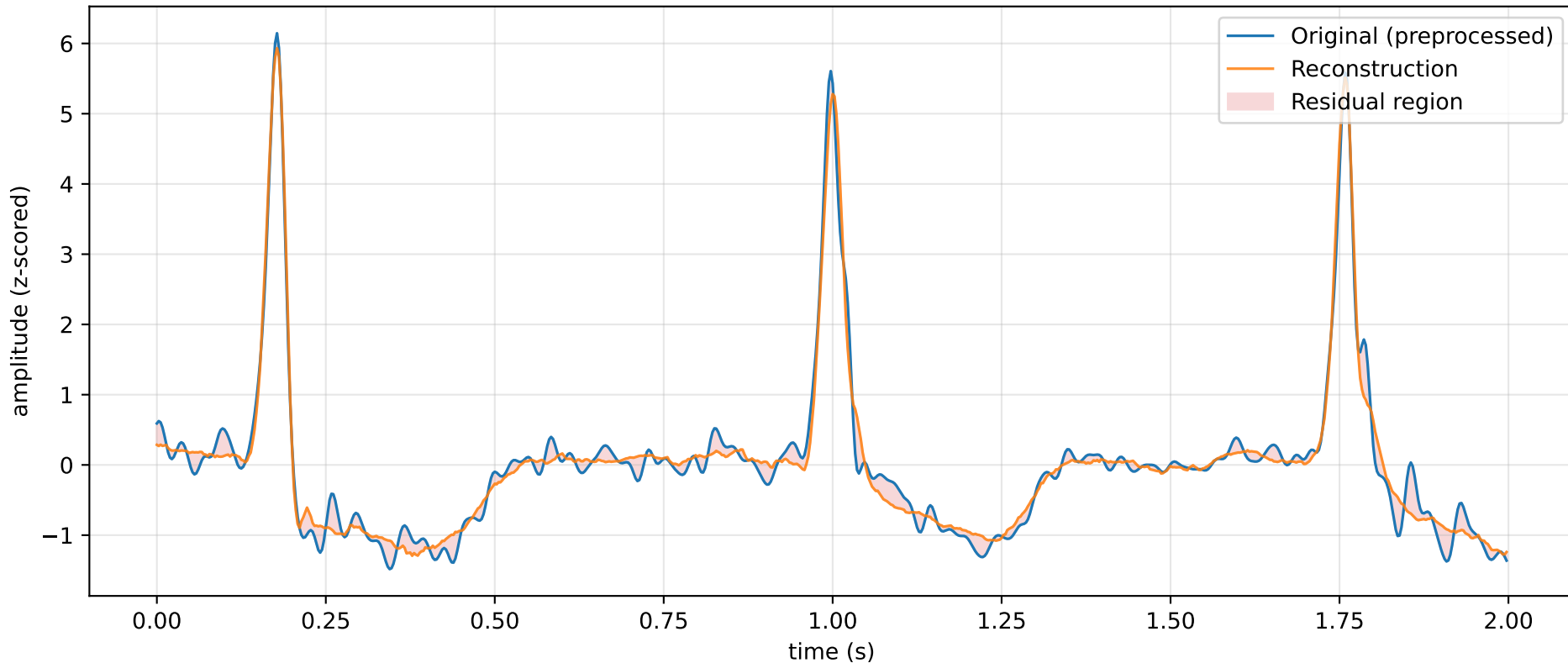
### Beat #36 - TN: ground truth = Normal beat (N)



Source : rec 101, sample 303862  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 0.0009  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

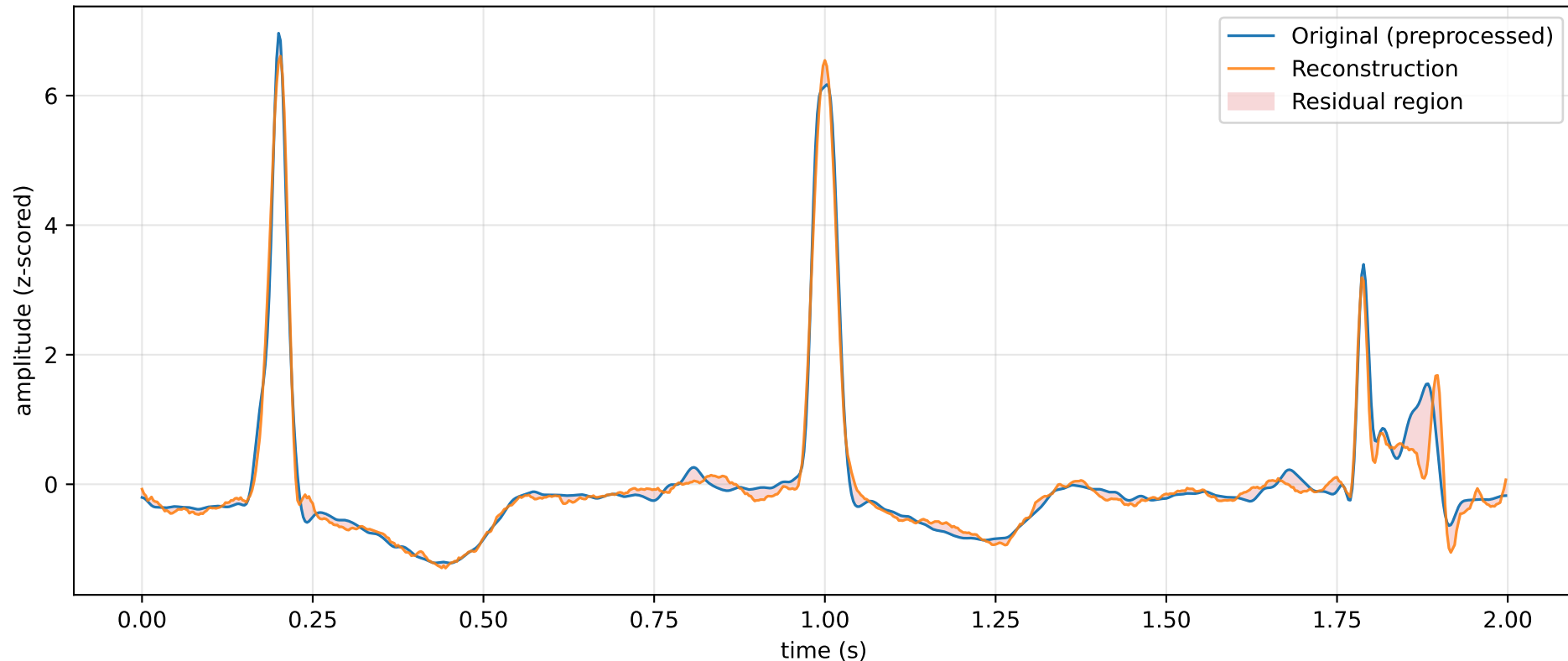
### Beat #37 - FN: ground truth = Fusion of paced and normal (f)



Source : rec 104, sample 106625  
Beat symbol : f (Fusion of paced and normal)  
Ground truth : ANOMALY  
Residual : 0.0433  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

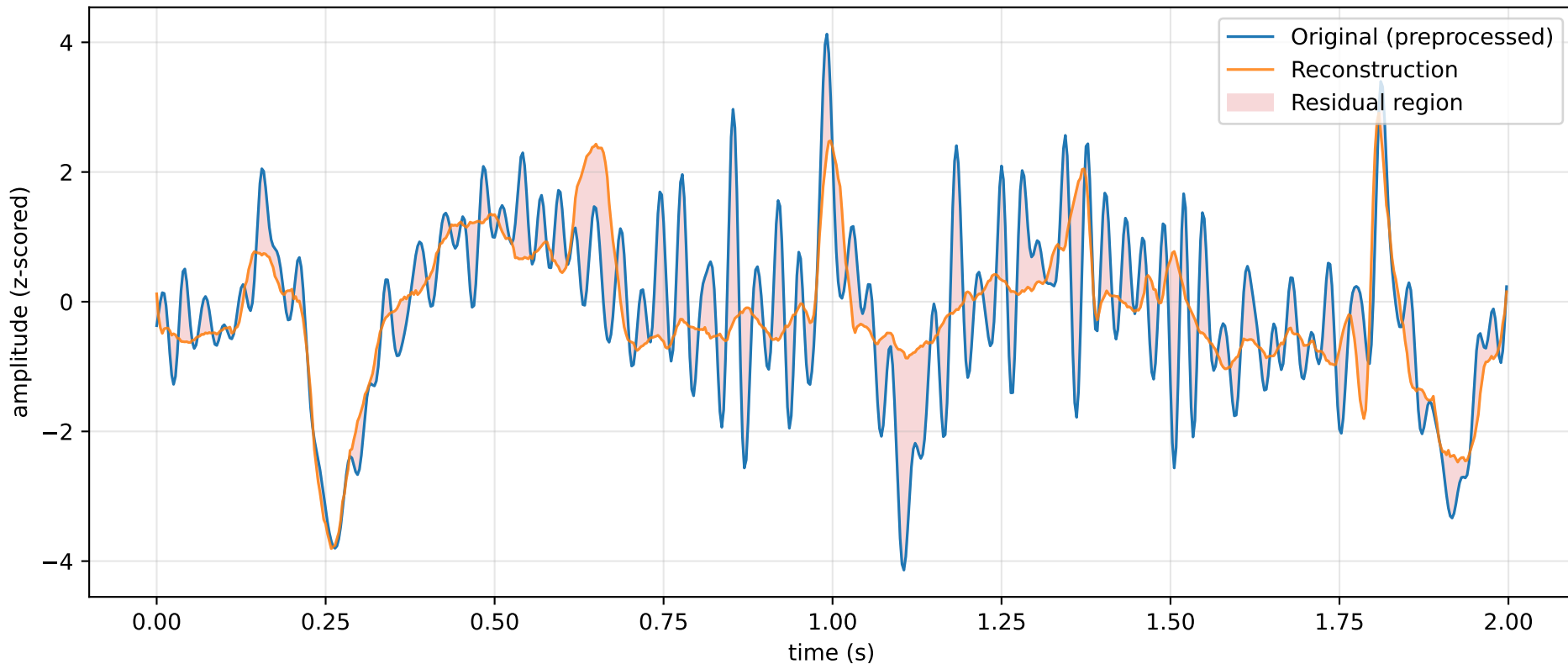
### Beat #38 - FN: ground truth = Fusion of paced and normal (f)



Source : rec 104, sample 284556  
Beat symbol : f (Fusion of paced and normal)  
Ground truth : ANOMALY  
Residual : 0.0433  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

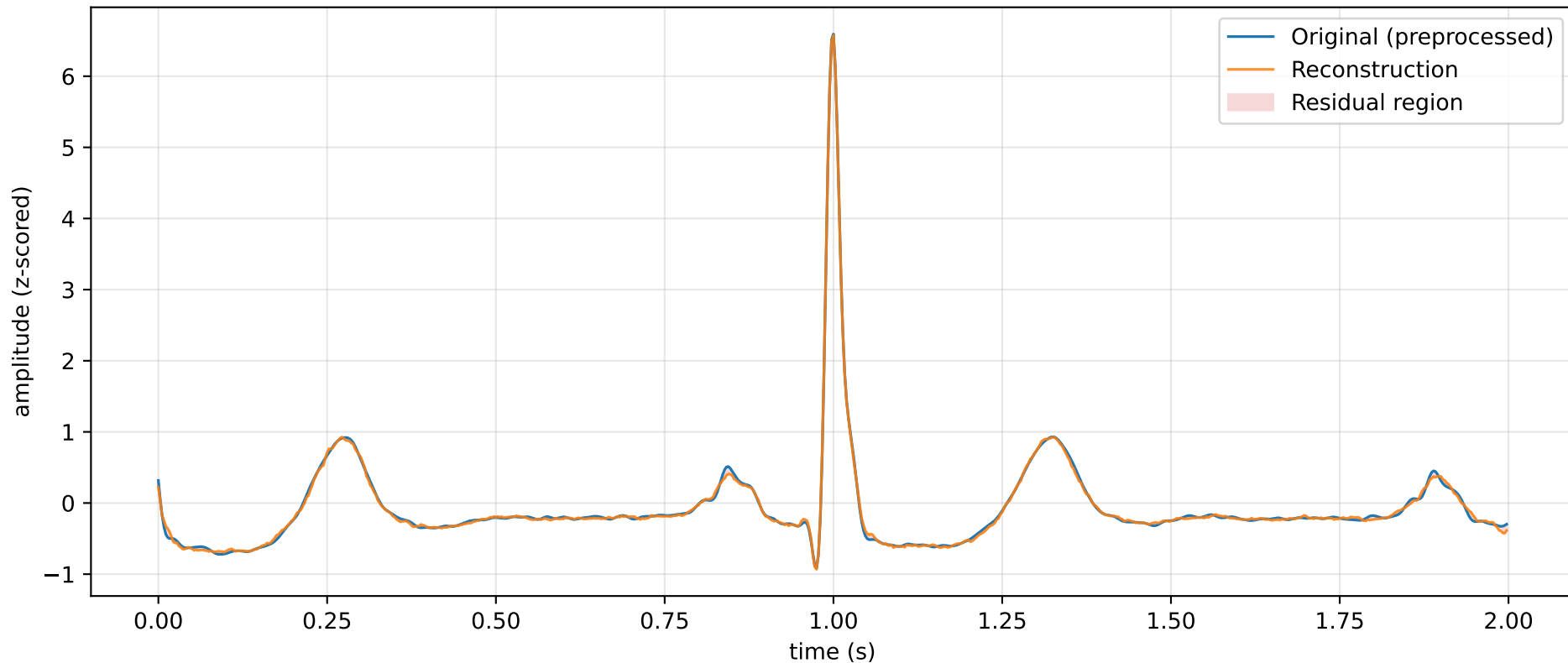
### Beat #39 - TP: ground truth = Paced beat (/)



Source : rec 104, sample 47938  
Beat symbol : / (Paced beat)  
Ground truth : ANOMALY  
Residual : 0.9589  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

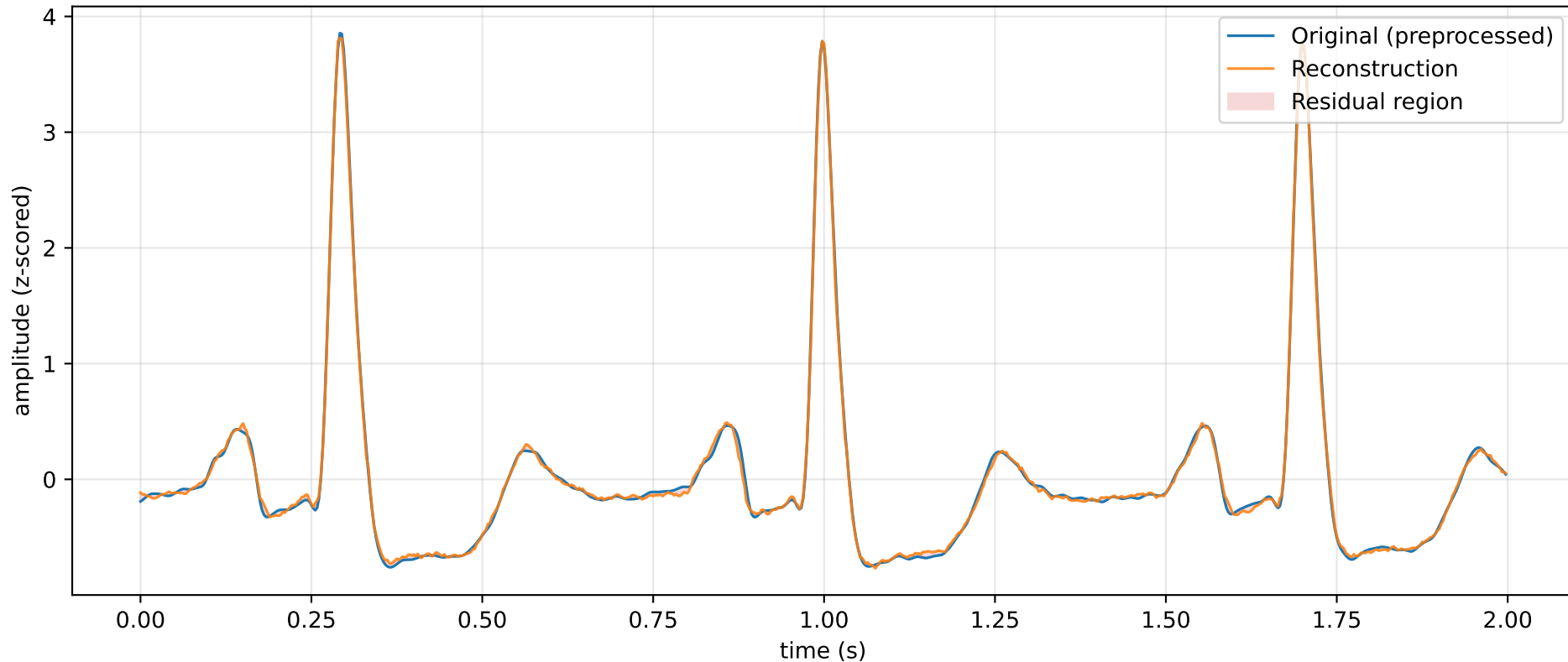
# Beat #40 - TN: ground truth = Normal beat (N)



Source : rec 101, sample 331197  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 0.0009  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

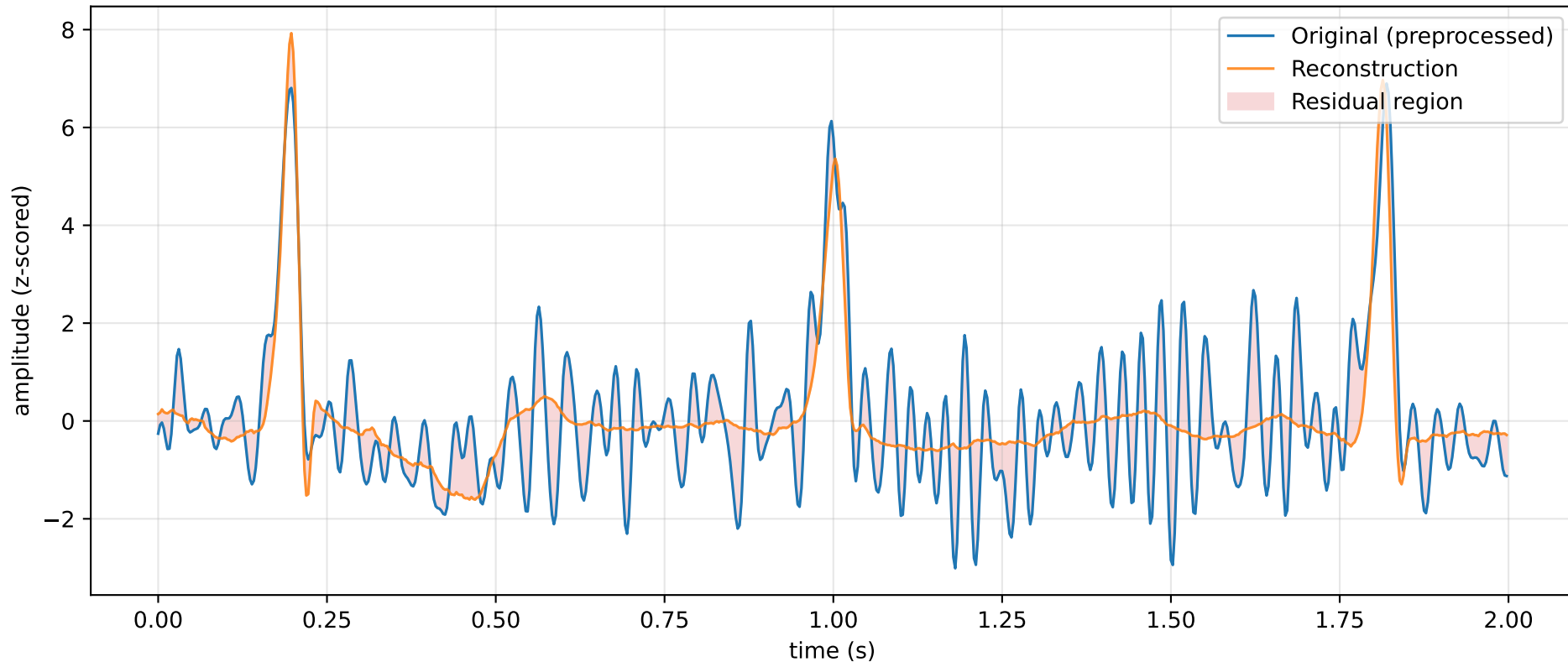
# Beat #41 - TN: ground truth = Normal beat (N)



Source : rec 105, sample 20262  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 0.0009  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

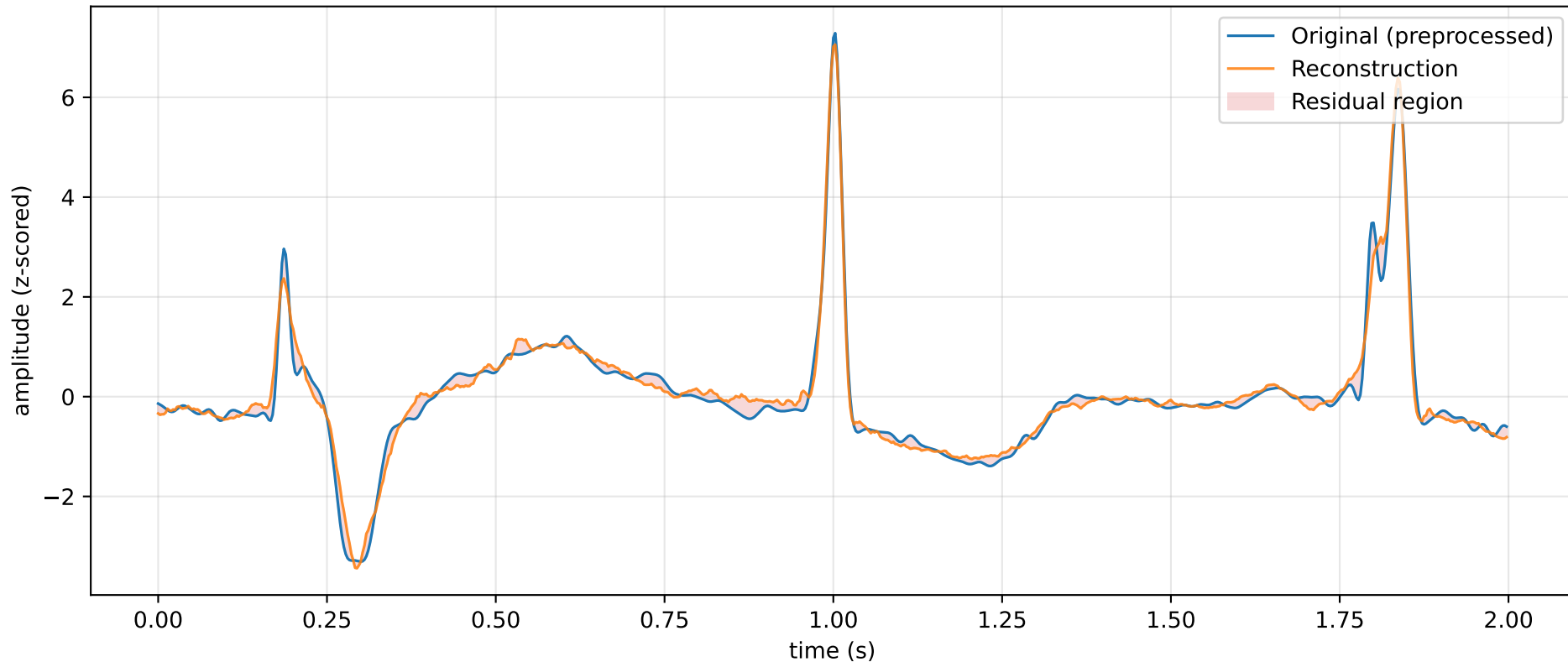
# Beat #42 - TP: ground truth = Unclassifiable (Q)



Source : rec 104, sample 111856  
Beat symbol : Q (Unclassifiable)  
Ground truth : ANOMALY  
Residual : 1.0223  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

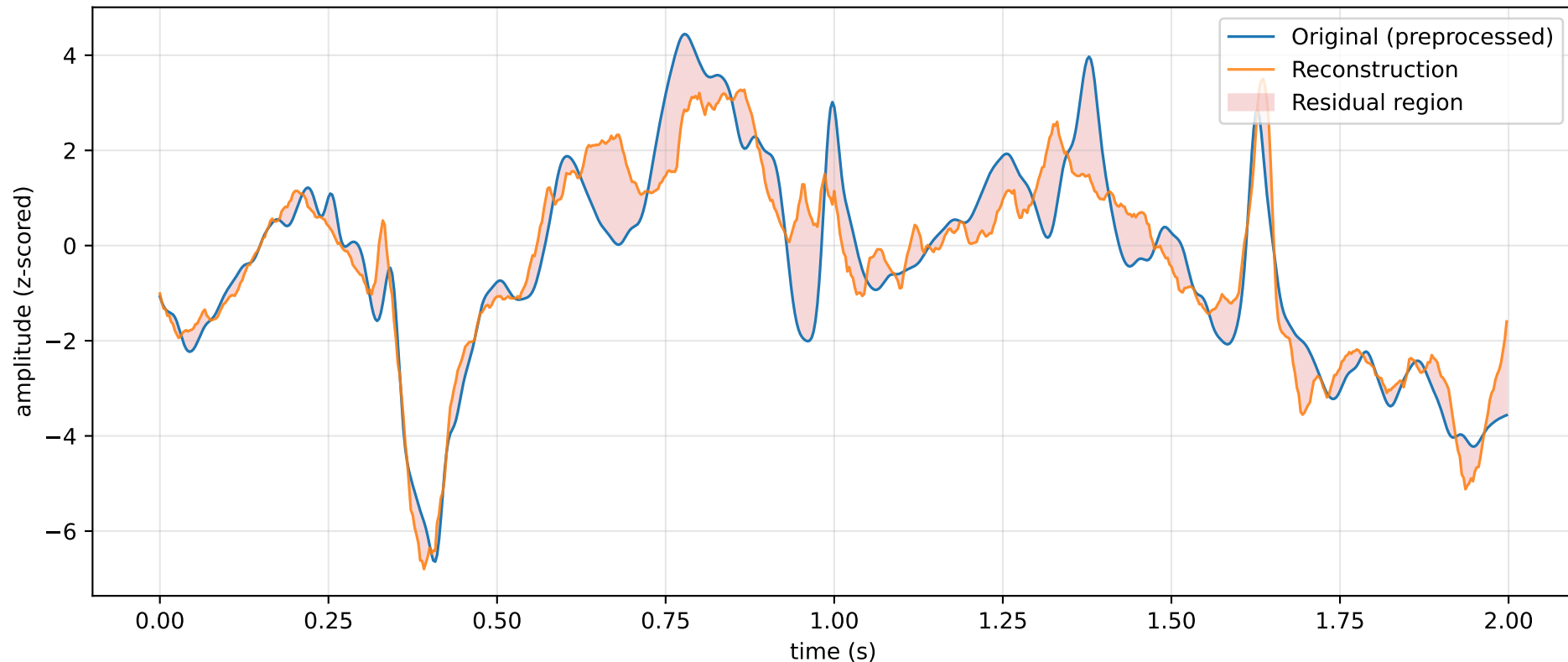
### Beat #43 - FN: ground truth = Fusion of paced and normal (f)



Source : rec 104, sample 15816  
Beat symbol : f (Fusion of paced and normal)  
Ground truth : ANOMALY  
Residual : 0.0432  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

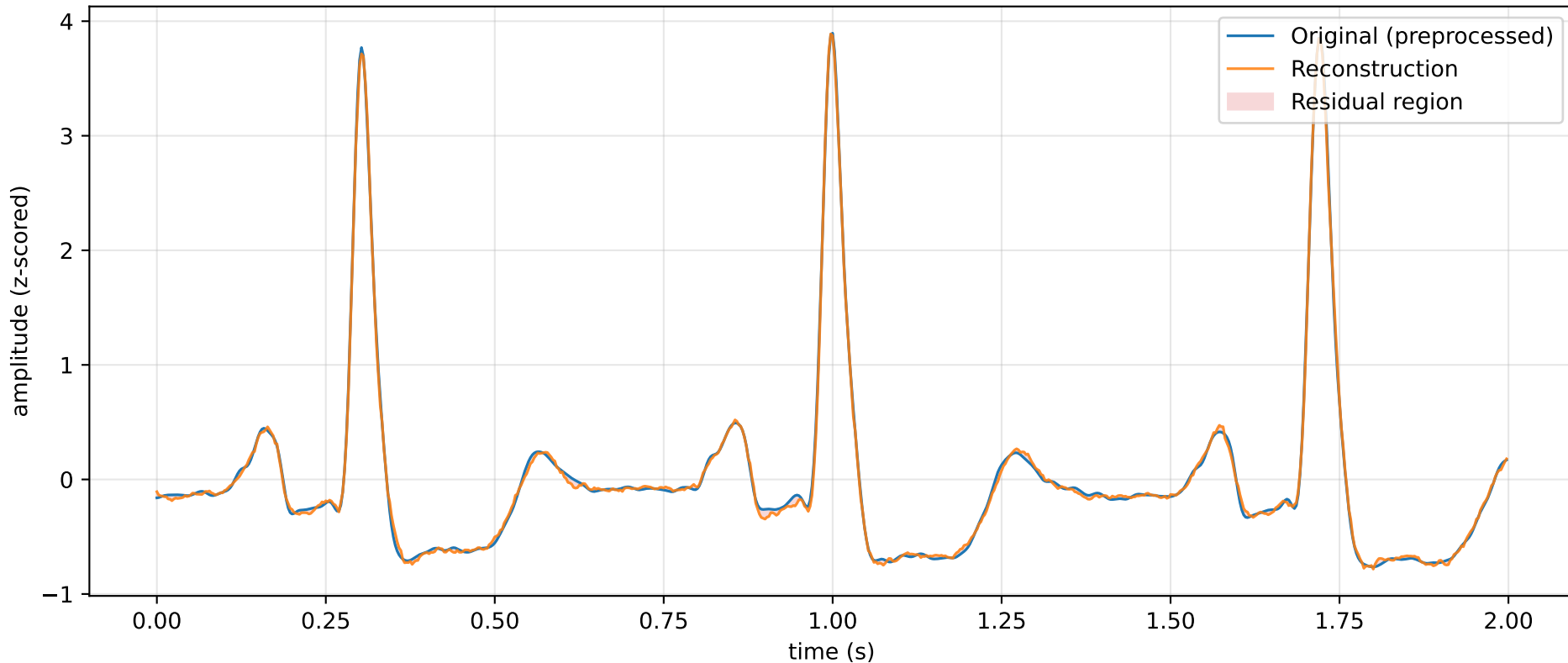
### Beat #44 - FP: ground truth = Normal beat (N)



Source : rec 105, sample 439248  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 0.7986  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

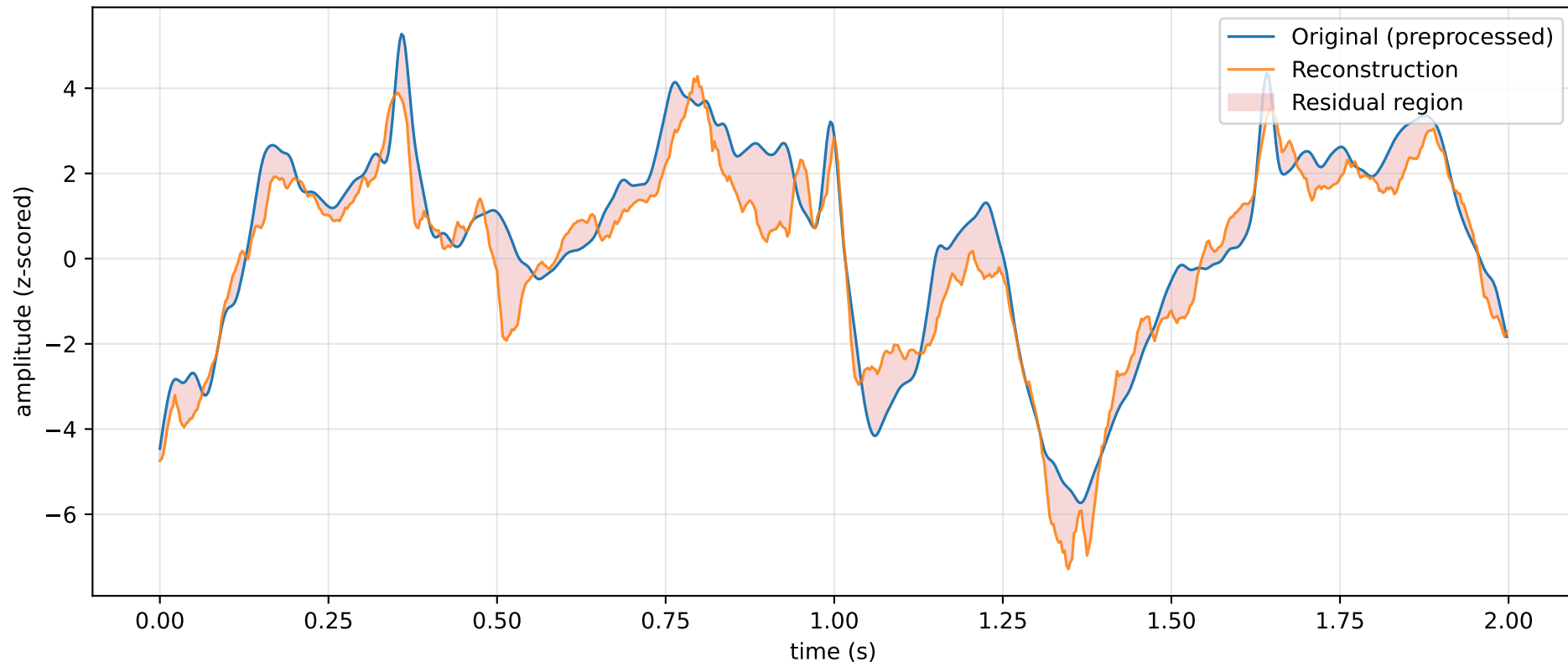
### Beat #45 - TN: ground truth = Normal beat (N)



Source : rec 105, sample 7246  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 0.0009  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

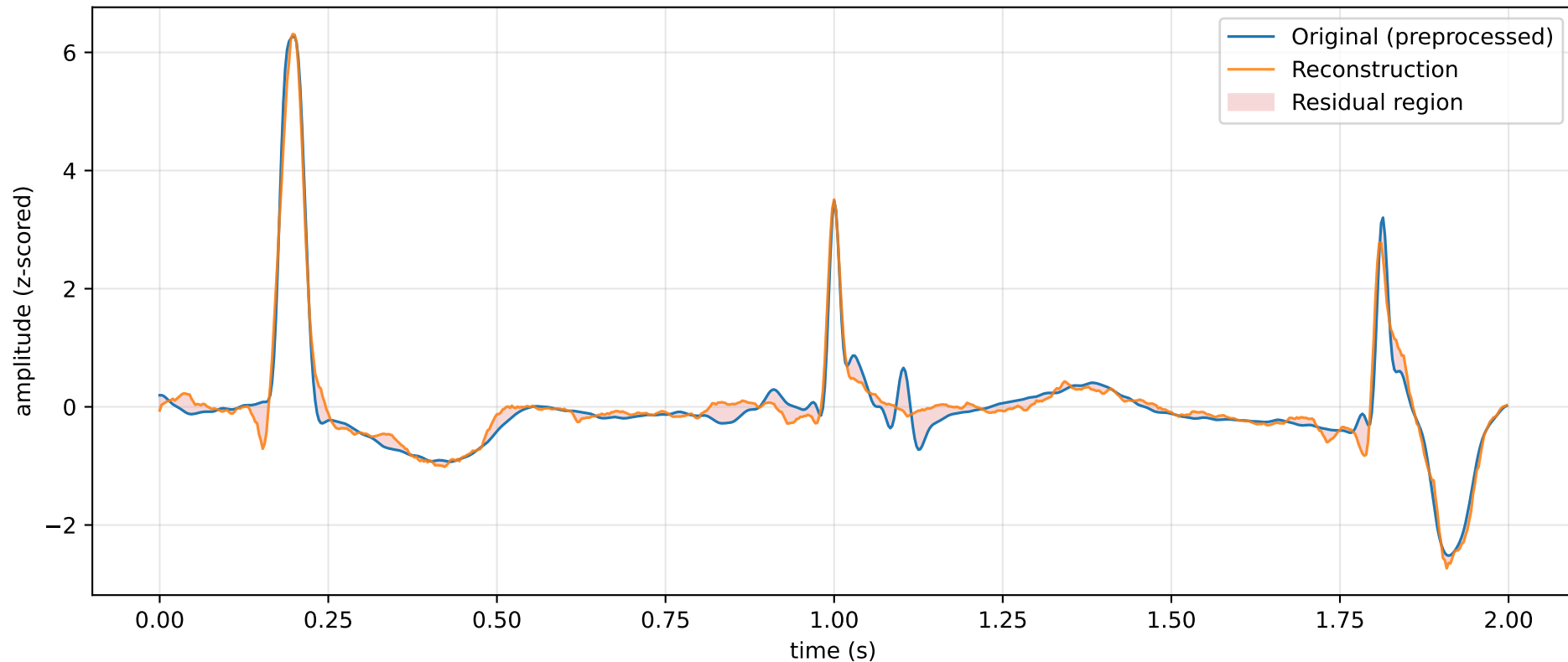
## Beat #46 - FP: ground truth = Normal beat (N)



Source : rec 105, sample 443710  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 0.7294  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**

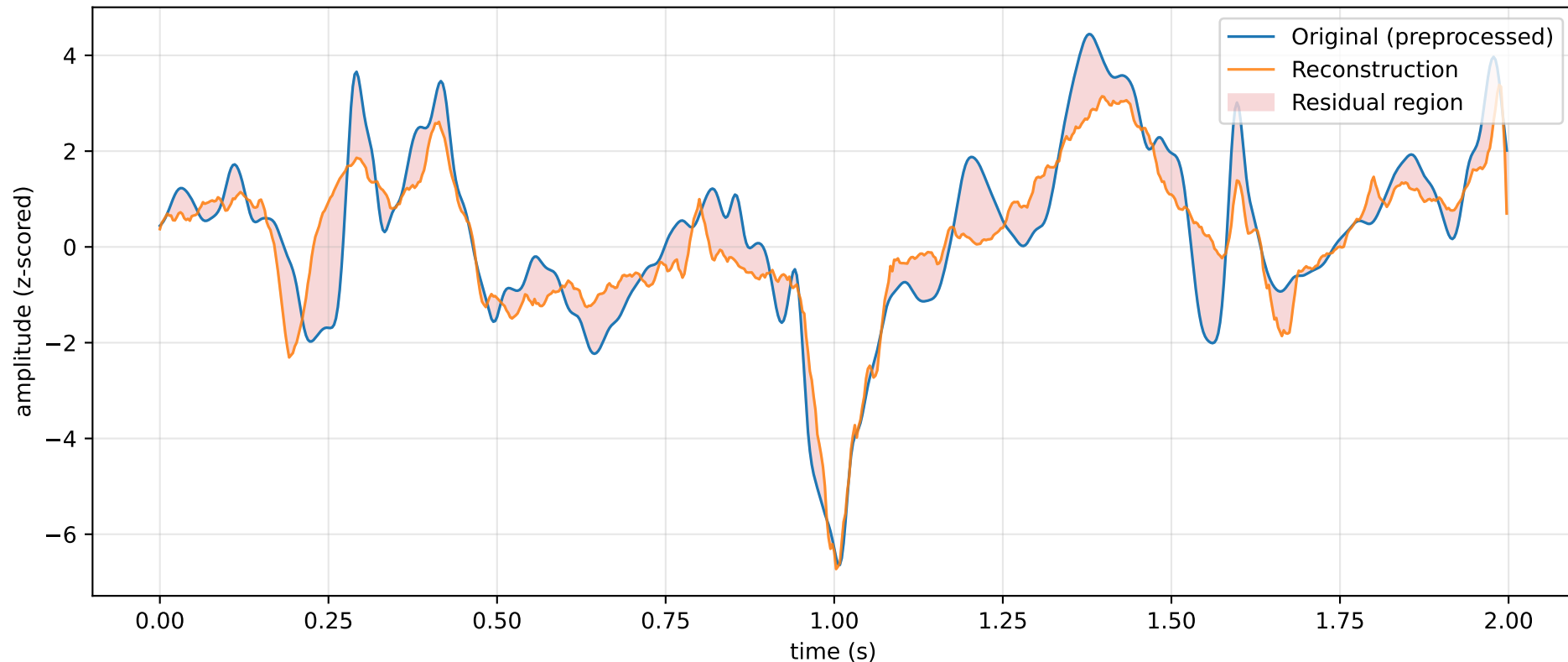
### Beat #47 - FN: ground truth = Fusion of paced and normal (f)



Source : rec 104, sample 571806  
Beat symbol : f (Fusion of paced and normal)  
Ground truth : ANOMALY  
Residual : 0.0432  
Threshold : 0.0434  
Model says : NORMAL

**NORMAL**

### Beat #48 - FP: ground truth = Normal beat (N)



Source : rec 105, sample 439032  
Beat symbol : N (Normal beat)  
Ground truth : NORMAL  
Residual : 0.7427  
Threshold : 0.0434  
Model says : ANOMALY

**ANOMALY**